

Reasoning about Data Structures with CDSAT¹

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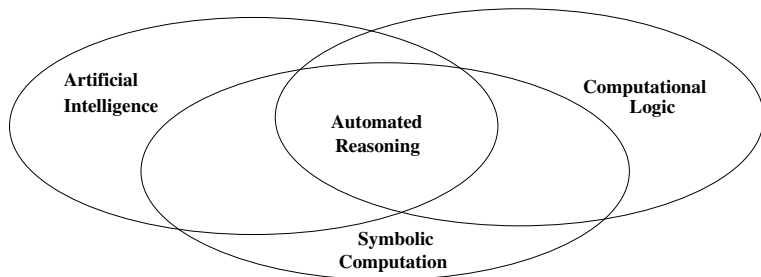
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(Subsumes a 2h invited tutorial entitled “The CDSAT Paradigm for Theory Combination in SMT,”
21st Int. Conf. on Computability in Europe (CiE), Lisbon, Portugal, EU, 15 and 17 July 2025,
and a 3h invited talk of the same title, CEA, Paris, France, EU, 15 October 2025)

¹Based on joint work with S. Graham-Lengrand and N. Shankar

- ▶ Introduction and motivation
- ▶ Conflict-driven reasoning and CDSAT in a nutshell
- ▶ Propositional conflict-driven reasoning: CDSAT and CDCL
- ▶ Combination of theories: CDSAT and Nelson-Oppen
- ▶ Conflict-driven reasoning in one theory: LRA procedures

What is automated reasoning (AR)



- ▶ AR: enable computers to reason ... in their own way
- ▶ AR is AI: logic-based, symbolic AI
- ▶ AR is SC: logico-deductive, symbolic reasoning
- ▶ AR is CL: use computing to perform logical reasoning

What automated reasoning does

- ▶ Design and implementation of computer programs that reason
 - ▶ To **solve** problems formulated as
 - ▶ **Validity** or **satisfiability** queries in a logic or a theory
 - ▶ Using **inference** and **search**
-
- ▶ **Valid**: true in all circumstances
 - ▶ **Satisfiable**: true in at least one circumstance

Why automated reasoning at a verification school

Applications of automated reasoning:

- ▶ AR is embedded in tools for the **verification**, synthesis, and optimization of software
- ▶ AR gets combined with:
 - ▶ **Testing**: AR for test generation
 - ▶ **Model checking**: AR for bounded model checking and abstraction in counter-example guided abstraction refinement
 - ▶ **Static analysis**: AR for invariant synthesis
- ▶ AR is also applied in deductive knowledge bases, computer mathematics, mathematical libraries, education, ...
- ▶ Could the integration of AR and ML lead to a better AI?

Objectives of verification/synthesis:

- ▶ Classic:
 - ▶ Partial correctness, termination
 - ▶ Correct-by-construction software
- ▶ More recent:
 - ▶ Provable privacy
 - ▶ Distributed systems/protocols
 - ▶ Randomized algorithms
 - ▶ Probabilistic programs

Automated reasoning and deductive verification

Objectives: partial correctness, termination

- ▶ Program state: **assignment** to program variables
- ▶ **Annotation**: formula where the only free (i.e., not quantified) variables are program variables
- ▶ **Annotated program**: code interspersed with annotations
- ▶ When the execution reaches the annotation, the program state should satisfy the annotation:
the annotation is an **invariant**
- ▶ Key analogy: a program state as a **model**

Automated reasoning and deductive verification

- ▶ **Verifying compiler**: takes annotated program and generates executable + **verification conditions** (VCs)
- ▶ If the VC's are valid, the program satisfies the annotations
- ▶ Automated reasoner: takes each VC and generates either **proof of validity** or **counter-model**
- ▶ **Proof**: certificate of correctness
- ▶ **Counter-model**: assignment to program variables useful for debugging

The automated reasoning problem

- ▶ Given a set H of assumptions and a conjecture φ

- ▶ Problem:

$H \models \varphi$: is φ a logical consequence of H ?

$\models (H \supset \varphi)$: is $H \supset \varphi$ valid?

$H \cup \{\neg\varphi\} \models \perp$: is $H \cup \{\neg\varphi\}$ unsatisfiable?

- ▶ Solution:

Either a **proof** of the unsatisfiability of $H \cup \{\neg\varphi\}$

Or a **model** of $H \cup \{\neg\varphi\}$: **counter-model** of φ

Discharging verification conditions is an instance of the AR problem

The automated reasoning problem in these lectures

- ▶ Satisfiability modulo theory: **defined** symbols in queries
Not many theories, but more than one: **combination**
Verification conditions: e.g., arithmetic and data structures
- ▶ Fragments where satisfiability is **decidable**: decision procedure
Almost a requirement for verification
- ▶ **Quantifier-free** and **clausal** input problems
Quantifier-free: a limitation, but there is progress
Clauses: standard machine language for logic
- ▶ Focus on **conflict-driven** reasoning: key ingredient
in SAT and SMT solvers (not the only one)

Examples: clauses involving theory-defined symbols

- ▶ Propositional logic (the **Boolean** theory):
 $\{\bar{A} \vee B, \bar{A} \vee C \vee E, \bar{B} \vee D, \bar{C} \vee \bar{D}, A \vee \bar{B} \vee E, B \vee \bar{C}, F \vee \bar{E}\}$
- ▶ Linear integer arithmetic (**LIA**) and Equality with Uninterpreted Functions (**EUF** or **UF**):
 $\{x \leq y, y \leq (x + g(x)), p(h(x) - h(y)), \neg p(0), g(x) \simeq 0\}$
- ▶ **Bool** and linear rational arithmetic (**LRA**):
 $\{x < y, x < z, (y < w) \vee (z < w), w < x\}$
- ▶ **Bool**, **LRA**, and Arrays (**Arr**):
 $(i \neq j) \vee (\text{select}(\text{store}(a, i, v), j) < \text{select}(a, j))$
 $(\text{select}(a, j) - \text{select}(a, k)) \simeq 0$
 $(\text{select}(\text{store}(a, i, v), j) \not< \text{select}(a, j)) \vee$
 $(\text{select}(a, j) + \text{select}(a, k) \simeq v)$

Conflict-driven reasoning

- ▶ Decision procedure for satisfiability of a set of clauses
- ▶ **Search** for a model: build candidate, propose assignments, amend to solve conflict
- ▶ Perform **inferences** to
 - ▶ Simplify the problem
 - ▶ Propagate consequences of assignments
 - ▶ Explain conflict btw candidate model and clauses
 - ▶ Prove unsatisfiability
- ▶ **Search** and **Inference** guide each other:
 - ▶ **Search** focuses **Inference** on conflicts which may facilitate proving unsatisfiability
 - ▶ **Inference** allows **search** to escape dead-end's

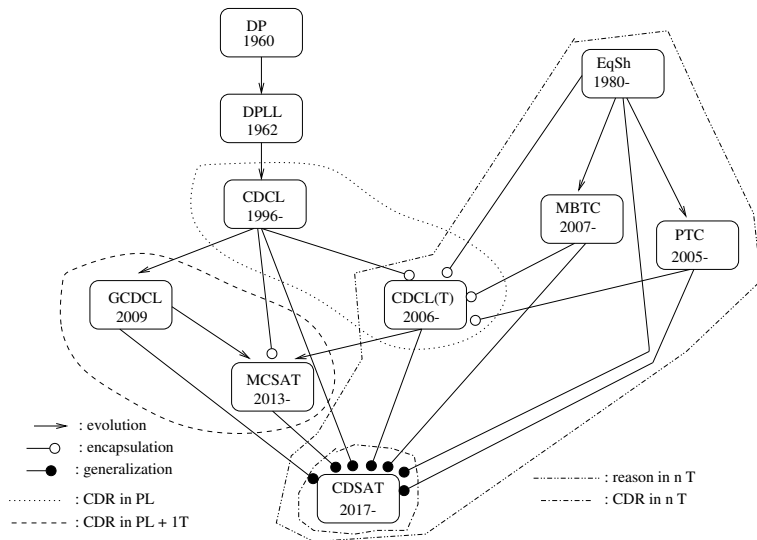
Conflict-driven reasoning

- ▶ **Search** for a model:
 - ▶ Decide **assignments** of values to terms
 - ▶ **Propagate** consequences of assignments (“inexpensive” inferences)
 - ▶ **Conflict**: contradiction
- ▶ Either reach unsatisfiability or solve conflict:
 - ▶ **Explain** conflict by “expensive” **inferences** (steps towards a possible refutation)
 - ▶ **Learn** generated **lemma** which excludes current assignment and avoids hitting same conflict
 - ▶ Solve conflict by amending assignment to satisfy lemma

Plan of these lectures

- ▶ Present the conflict-driven method **CDSAT** (**Conflict-Driven Satisfiability**) for satisfiability modulo a **union** of theories [MPB, S. Graham-Lengrand, N. Shankar: 2017-]
- ▶ Use it as a lense to revisit several SMT methods that CDSAT generalizes, introducing gradually most CDSAT features
- ▶ Illustrate an application of CDSAT to reasoning about novel theories of **arrays**, **maps**, and **vectors** with **abstract domain** that increase expressiveness capturing realistic features of these data structures

The big picture: conflict-driven reasoning procedures



CDSAT: conflict-driven reasoning in theory unions

- ▶ Decide satisfiability in a **theory union** $\mathcal{T} = \bigcup_{k=1}^n \mathcal{T}_k$
- ▶ **Predicate-sharing** theories
Disjoint theories if $\simeq$ is the only shared predicate
- ▶ SMA (Satisfiability Modulo Theory and Assignment):
input includes **initial assignment**
 - ▶ **Boolean assignment**: $L \leftarrow \text{true}$ (**Boolean value**)
 - ▶ **First-order assignment**: $x \leftarrow 3$ (**non-Boolean value**)
 - ▶ Relevant for parallelization, model interpolation, quantified satisfiability (**QSMA**), hybridization of local search and complete SMT methods
- ▶ Answer **sat** if there exists satisfying assignment including initial one, **unsat** otherwise

CDSAT: conflict-driven reasoning in theory unions

- ▶ Transition system: transition rules (e.g., **Decide**, **Deduce**)
- ▶ Coordinates **theory modules**:
theory inference system + **finite local basis**
- ▶ Offers conflict-driven control accommodating also non-conflict-driven procedures (**black-box** modules)
- ▶ The modules collaborate as **peers**
on a **shared trail** Γ containing the current assignment
- ▶ Each module offers **decisions** and **deductions**
propagation, **simplification**, **conflict detection**, **conflict explanation**
- ▶ **Sound**, **complete**, **terminating** under suitable hypotheses
 - ▶ **Finite global basis** from the local ones for **termination**

Assignments take center stage

- ▶ Assignments of **values** to **terms**:
 $(x > 1) \leftarrow \text{false}, ((x > 1) \vee (y < 0)) \leftarrow \text{true},$
 $(\text{store}(a, i, v) \simeq b) \leftarrow \text{true}, y \leftarrow \sqrt{2}, \text{select}(a, j) \leftarrow 3$
- ▶ For each pair term and value have the same sort
- ▶ Boolean assignments are routinely abbreviated:
 $(x > 1) \vee (y < 0)$ for $((x > 1) \vee (y < 0)) \leftarrow \text{true}$
 $\overline{x > 1}$ for $(x > 1) \leftarrow \text{false}$
- ▶ Formulas are **Boolean** terms (sort prop)
- ▶ **Plausible** assignment: does not contain $L \leftarrow \text{true}$ and $L \leftarrow \text{false}$

Assignments take center stage

- ▶ **Terms** and **values** are kept separate:
term only on the left, **value** only on the right of an assignment
- ▶ $\text{select}(a, j) \leftarrow 3$ (assignment) cannot be replaced by
 $\text{select}(a, j) \simeq 3$ (literal)
and we cannot write $\text{select}(a, 3)$:
a value is not a term, is not in the signature
- ▶ What are **values**?

Theory extension

- ▶ Theory $\mathcal{T}$ with signature Σ
- ▶ **Theory extension** $\mathcal{T}^+$ with signature Σ^+ :
 - ▶ Add new constant symbols (and possibly axioms)
 - ▶ E.g.: add a constant symbol for every number (integers, rationals, algebraic reals)
 $\sqrt{2}$ is a constant symbol interpreted as $\sqrt{2}$
 - ▶ All $\mathcal{T}$ have sort prop: all $\mathcal{T}^+$ add true and false
- ▶ **Conservative** extension: for Σ -formulas
 - ▶ $\mathcal{T}$ -satisfiable implies $\mathcal{T}^+$ -satisfiable, or, equivalently,
 - ▶ $\mathcal{T}^+$ -unsatisfiable implies $\mathcal{T}$ -unsatisfiable

Theory extensions provide values

- ▶ From theory $\mathcal{T}$ to theory extension $\mathcal{T}^+$
- ▶ **Values** in assignments are the constant symbols added by $\mathcal{T}^+$: they are called **$\mathcal{T}$ -values**
- ▶ An assignment $\{x_1 \leftarrow c_1, \dots, x_m \leftarrow c_m\}$ is a **$\mathcal{T}$ -assignment** if for all i , $1 \leq i \leq m$, c_i is a **$\mathcal{T}$ -value**

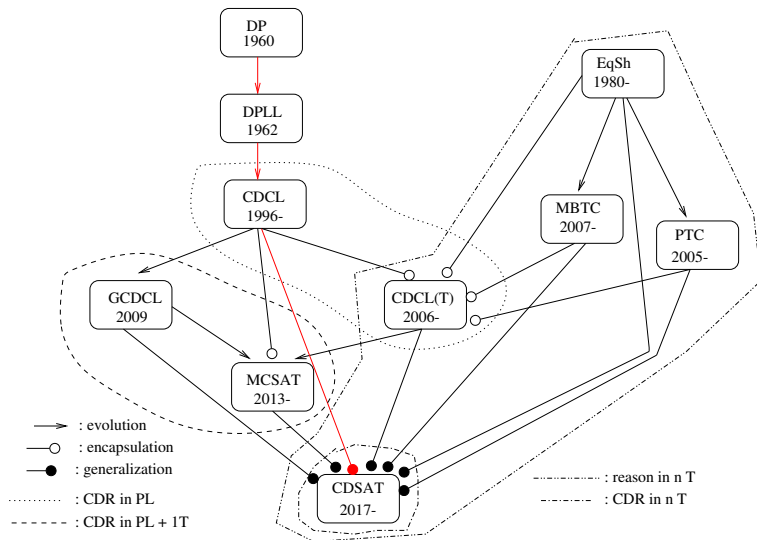
Two kinds of theory extension

- ▶ **Trivial**: adds **only** true and false with axioms $\{\text{true}, \neg\text{false}\}$
- ▶ **Countable**: adds true and false with axioms $\{\text{true}, \neg\text{false}\}$ for sort prop, and a countably infinite set of **$\mathcal{T}$ -values** for each sort s other than prop
- ▶ A countable extension may or may not add axioms, e.g.:
 - ▶ Equality (**EU F^+**): no axioms, the **$\mathcal{T}$ -values** act as labels of congruence classes
 - ▶ **NRA $^+$** : the sentences true in the intended model of the reals interpreting every **NRA-value** as itself (e.g., $\sqrt{2}$ with $\sqrt{2} \cdot \sqrt{2} \simeq 2$)

Theory modules $\mathcal{I}_1, \dots, \mathcal{I}_n$ for theories $\mathcal{T}_1, \dots, \mathcal{T}_n$

- ▶ $\mathcal{I}_k$ is a set of inference rules $J \vdash_k L$ where
 - ▶ J is a $\mathcal{T}_k$ -assignment: may contain first-order assignments (a $\mathcal{T}_k$ -assignment assigns $\mathcal{T}_k$ -values)
 - ▶ L is a singleton **Boolean** assignment
- ▶ A function **basis_k** from sets of terms to sets of terms called **local basis**:
 - ▶ Given a set **X** of terms (e.g., the input) terms, **basis_k(X)** is the **finite** set of all terms that the $\mathcal{I}_k$ inferences can generate

The big picture: propositional reasoning



Propositional satisfiability

- ▶ DP [Davis, Putnam: JACM 1960]:
 - ▶ **Resolution** ($C \vee L$ and $D \vee \bar{L}$ resolve to generate $C \vee D$)
 - ▶ **Subsumption** (L subsumes $C \vee L$, and D subsumes $D \vee \bar{L}$)
- ▶ DPLL [Davis, Putnam, Logeman, Loveland: CACM 1962]:
 - ▶ **Backtracking search** for a model
partial model represented by Boolean assignments
on a **trail** Γ (stack)
 - ▶ Resolution replaced by **splitting** on L and $\bar{L}$
 - ▶ **Unit propagation**: unit subsumption + unit resolution:
if L on Γ , delete $C \vee L$, and replace $D \vee \bar{L}$ with D
 - ▶ **Conflict** (e.g., $\{P, \bar{P}\}$): **backtrack** last guess
(e.g., from L to $\bar{L}$) or **fail** and return unsat if none
- ▶ CDCL [Marques Silva, Sakallah: ICCAD 1996, IEEE TOC 1999]

CDCL: Conflict-Driven Clause Learning

- ▶ **Decision** replaces the splitting of DPLL:
add L to trail Γ provided $L \notin \Gamma$ and $\bar{L} \notin \Gamma$
- ▶ Every decision opens new level on trail Γ which is a stack
- ▶ **Unit propagation** detects
 - ▶ **Implied literal** L with **justification** $C = L_1 \vee \dots \vee L_k \vee L$
if $\bar{L}_i \in \Gamma$ ($1 \leq i \leq k$) and adds L to current level
 - ▶ **Conflict clause** $D = Q_1 \vee \dots \vee Q_n$ if $\bar{Q}_i \in \Gamma$ ($1 \leq i \leq n$)
- ▶ Conflict at level 0 (no decision to flip):
fail and return unsat
- ▶ Otherwise: **explain** and/or **solve conflict**

CDCL: Conflict-Driven Clause Learning

- ▶ Explain conflict by resolving conflict clause $D = Q_1 \vee \dots \vee Q_n$ and justification of $\overline{Q_i}$
- ▶ The generated resolvent is still a conflict clause
- ▶ Learn a resolvent C as a lemma
- ▶ A heuristic determines how many resolution steps and which lemma
- ▶ Conflict-driven backjumping replaces the backtracking of DPLL: backjump away from conflict to a state that satisfies C (backjump clause)

The first assertion clause heuristic

- ▶ Resolve until the conflict clause is an **assertion clause**
- ▶ Assertion clause: a conflict clause $C = L_1 \vee \dots \vee L_k \vee L$ where only L is falsified by the current level (1UIP)
- ▶ L is the **assertion literal**
- ▶ Learn C
- ▶ Backjump to the smallest level such that $\bar{L}_i \in \Gamma$ ($1 \leq i \leq k$) and L undefined
- ▶ L is an implied literal with justification C

CDCL procedure: example

$S = \{\bar{A} \vee B, \bar{A} \vee C \vee E, \bar{B} \vee D, \bar{C} \vee \bar{D}, A \vee \bar{B} \vee E, B \vee \bar{C}, F \vee \bar{E}\}$
subset of input

1. A **decision** adds $\bar{F}$ to trail Γ (opening level n)
2. **Unit propagation** adds $\bar{E}$ with **justification** $F \vee \bar{E}$ (level n)
3. Two more decisions unrelated to S (levels $n+1$ and $n+2$)
4. Another decision adds A to Γ (level $n+3$)
5. Unit propagation adds B with justification $\bar{A} \vee B$
 C with justification $\bar{A} \vee C \vee E$
 D with justification $\bar{B} \vee D$ (all on level $n+3$)

CDCL procedure: example

6. **Conflict:** $\overline{C} \vee \overline{D}$ is **conflict clause**
7. Why is $\overline{D}$ false? Because D is true with justification $\overline{B} \vee D$
Resolve $\overline{C} \vee \overline{D}$ and $\overline{B} \vee D$ to yield $\overline{B} \vee \overline{C}$
(still a conflict clause)
8. Why is $\overline{C}$ false? Because C is true with justification $\overline{A} \vee C \vee E$
Resolve $\overline{B} \vee \overline{C}$ and $\overline{A} \vee C \vee E$ to yield $\overline{B} \vee \overline{A} \vee E$
(still a conflict clause)
9. Why is $\overline{B}$ false? Because B is true with justification $\overline{A} \vee B$
Resolve $\overline{B} \vee \overline{A} \vee E$ and $\overline{A} \vee B$ to yield $\overline{A} \vee E$
 $\overline{A} \vee E$: only $\overline{A}$ is falsified by the current level $n + 3$

CDCL procedure: example

10. **Learn** $\bar{A} \vee E$: no model with A and $\bar{E}$
11. **Backjump** to the smallest level where $\bar{A}$ is undefined and E is still false: level n
Add $\bar{A}$ to level n with justification $\bar{A} \vee E$
12. Unit propagation adds
 $\bar{B}$ with justification $A \vee \bar{B} \vee E$
 $\bar{C}$ with justification $B \vee \bar{C}$ (both on level n)
 Γ contains $\{\bar{E}, \bar{A}, \bar{B}, \bar{C}\}$ which is a model of S

CDSAT emulates CDCL if **Bool** is the only theory in the union $\mathcal{T}$

CDSAT module for theory Bool

- ▶ $\Sigma_{\text{Bool}} = \langle \{ \text{prop} \}, \{ \neg, \vee, \wedge, \simeq_{\text{prop}} \} \rangle$
- ▶ Theory extension Bool^+ is **trivial**
- ▶ **Evaluation**: $(L_1 \leftarrow b_1, \dots, L_m \leftarrow b_m) \vdash_{\text{Bool}} L \leftarrow b$
where each b_i and b is true or false
- ▶ $\neg$ -elim: $\neg L \vdash_{\text{Bool}} \bar{L}$ and $\neg \bar{L} \vdash_{\text{Bool}} L$
- ▶ $\wedge$ -elim: $L_1 \wedge \dots \wedge L_m \vdash_{\text{Bool}} L_i$ and $\overline{L_1 \vee \dots \vee L_m} \vdash_{\text{Bool}} \bar{L}_i$
- ▶ **Unit propagation**: $L_1 \vee \dots \vee L_m, \{\bar{L}_j \mid j \neq i\} \vdash_{\text{Bool}} L_i$ and $\overline{L_1 \wedge \dots \wedge L_m}, \{L_j \mid j \neq i\} \vdash_{\text{Bool}} \bar{L}_i$
- ▶ **basis_{Bool}(X)**: all subformulas of formulas in X
and all their disjunctions (for clause learning)

The CDSAT trail for the Boolean case

- ▶ The trail is a sequence of assignments: $\Gamma = L_0, \dots, L_m$
Boolean term L : L is $L \leftarrow \text{true}$, $\bar{L}$ is $L \leftarrow \text{false}$
Satisfaction of assignment: same as satisfaction of formula
- ▶ Input H_0 : written as an assignment $H_0 \subseteq \Gamma$
- ▶ Two kinds of assignments:
 - ▶ **Decision**: $?L$ (a guess)
 - ▶ **Justified assignment**: $H \vdash L$
justification H : a set of assignments that appear before L in Γ
Sound: $(H_0 \cup H) \models L$
- ▶ Input assignments: $\emptyset \vdash L$

The trail is organized in levels

$$\Gamma = L_0, \dots, L_m$$

- ▶ Every **decision** $?L$ opens a new **level**:
 $\text{level}_\Gamma(L_i) = 1 + \max\{\text{level}_\Gamma(L_j) \mid j < i\}$ if L_i is a decision $?L$
- ▶ A **justified assignment** $H \vdash L$ has the **level** of its **justification** H :
 $\text{level}_\Gamma(H) = 0$ if $H = \emptyset$ (e.g., input assignment)
 $\text{level}_\Gamma(H) = \max\{\text{level}_\Gamma(A) \mid A \in H\}$
 $H \vdash L$ has level 0 also when all elements in H have level 0

- ▶ **CDSAT-derivation**: a sequence of states where
 - ▶ The initial state is a trail Γ_0 containing only the input assignment H_0 , and
 - ▶ Every other state is generated from its predecessor by applying a CDSAT **transition rule**
- ▶ First kind of state: a trail Γ

Boolean decisions

Transition rule **Decide** adds to the trail a Boolean decision:

$$\text{Decide: } \Gamma \longrightarrow \Gamma, ?L$$

provided assignment L is **acceptable**:

$$L \notin \Gamma \text{ and } \bar{L} \notin \Gamma$$

which preserves **plausibility** of the assignment on the trail

- ▶ Transition rule **Deduce** adds to the trail a Boolean **justified assignment** $J \vdash L$:

$$\text{Deduce: } \Gamma \longrightarrow \Gamma, J \vdash L$$

if a $\mathcal{T}_k$ -module has inference $J \vdash_k L$ for $J \subseteq \Gamma$
and $L \notin \Gamma$ and $\bar{L} \notin \Gamma$

- ▶ **Deduce** covers **unit propagation** when $\mathcal{T}_k$ is **Bool**:

$$J = \{L_1 \vee \dots \vee L_k \vee L, \bar{L}_1, \dots, \bar{L}_k\}$$

Note that the **justification** is richer than in CDCL

- ▶ Given input assignment H_0 and trail Γ
- ▶ **Conflict**: an unsatisfiable subset of Γ
 $E \subseteq \Gamma, E \neq \emptyset, (H_0 \cup E) \models \perp$
- ▶ **Conflict detection**:
 - ▶ $J \subseteq \Gamma$
 - ▶ A $\mathcal{T}_k$ -module can infer $J \vdash_k L$ but $\bar{L} \in \Gamma$
 - ▶ Then $E = J \cup \{\bar{L}\}$ is a **conflict**
- ▶ The notion of conflict covers that of conflict clause:
 $E = \{Q_1 \vee \dots \vee Q_n, \bar{Q}_1, \dots, \bar{Q}_n\}$
conflict clause: negation of the **cube** $\bigwedge_{i=1}^n \bar{Q}_i$ in E
cube: set (conjunction) of literals

Conflict and failure

- ▶ Level of a **conflict** E : $\text{level}_\Gamma(E) = \max\{\text{level}_\Gamma(L) \mid L \in E\}$
- ▶ $\text{level}_\Gamma(E) = 0$:
 - since $E \neq \emptyset$, $\text{level}_\Gamma(L) = 0$ for all $L \in E$:
 - all $L \in E$ are input assignments or justified assignments derived from input assignments
 - no decision to undo, nowhere to backjump to
 - transition rule **Fail** returns **unsat**:

Fail: $\Gamma \longrightarrow \text{unsat}$

Conflict and conflict solving

- ▶ Level of a **conflict** E : $\text{level}_\Gamma(E) = \max\{\text{level}_\Gamma(L) \mid L \in E\}$
- ▶ $\text{level}_\Gamma(E) > 0$: transition rule **ConflictSolve** transfers control to the **conflict state rules** and returns the trail they produce:

$$\text{ConflictSolve}: \Gamma \longrightarrow \Gamma'$$

if $\langle \Gamma; E \rangle \Longrightarrow^* \Gamma'$

- ▶ $\longrightarrow$ for **trail rule** transition
 $\Longrightarrow$ for **conflict state rule** transition
- ▶ Second kind of state: **conflict state** $\langle \Gamma; E \rangle$

Conflict state rule Resolve

- ▶ Conflict state: $\langle \Gamma; E \uplus \{L\} \rangle$ and $H \vdash L \in \Gamma$
- ▶ Transition rule **Resolve** explains the conflict by replacing $H \vdash L$ with its justification H :

$$\text{Resolve: } \langle \Gamma; E \uplus \{L\} \rangle \Longrightarrow \langle \Gamma; E \cup H \rangle$$

- ▶ **Explanation:** why L ? Because of H
- ▶ $E \uplus H$ is still a conflict:
 $(H_0 \cup E \uplus \{L\}) \models \perp$ and $(H_0 \cup H) \models L$ imply
 $(H_0 \cup E \cup H) \models \perp$

Resolve covers propositional resolution

- ▶ Conflict $E_1 = \{Q_1 \vee \dots \vee Q_n, \overline{Q_1}, \dots, \overline{Q_n}\}$
Conflict clause in E_1 : $Q_1 \vee \dots \vee Q_n$
- ▶ Say $\overline{Q_1}$ has justification $\{\overline{Q_1} \vee C, \overline{C}\}$ where C is a disjunction of literals and $\overline{C}$ is the set of their flips
- ▶ **Resolve** applied to $\overline{Q_1}$ in E_1 yields
 $E_2 = \{Q_1 \vee \dots \vee Q_n, \overline{Q_1} \vee C, \overline{C}, \overline{Q_2}, \dots, \overline{Q_n}\}$
Conflict clause in E_2 : $C \vee Q_2 \vee \dots \vee Q_n$

Outstanding assignment and restriction of a trail

- ▶ Given conflict $E \uplus \{L\}$
assignment L is **outstanding** in $E \uplus \{L\}$
if $\text{level}_\Gamma(L) > \text{level}_\Gamma(E)$
- ▶ Given trail $\Gamma = L_0, \dots, L_m$
the **restriction** $\Gamma^{\leq m}$ to level m
contains all L_i such that $\text{level}_\Gamma(L_i) \leq m$
in the same order in which they appear in Γ

Conflict state rule Backjump

- ▶ Conflict state: $\langle \Gamma; E \uplus \{L\} \rangle$ where assignment L is **outstanding** in $E \uplus \{L\}$ and $\text{level}_\Gamma(E) = m$
- ▶ Transition rule **Backjump** solves the conflict by jumping back to level m and adding $_{E \vdash} \bar{L}$ to the trail:

$$\text{Backjump: } \langle \Gamma; E \uplus \{L\} \rangle \Longrightarrow \Gamma^{\leq m}, {}_{E \vdash} \bar{L}$$

- ▶ $(H_0 \cup E \uplus \{L\}) \models \perp$ implies $(H_0 \cup E) \models \bar{L}$:
justified assignment $_{E \vdash} \bar{L}$ is **sound**
- ▶ $\bar{L}$ is a **learned lemma**: no model can contain L as long as $H_0 \cup E$ is on the trail

First assertion clause heuristic with Backjump

- ▶ Apply **Resolve** until **conflict** has the form $E \uplus H$ where
 - ▶ $H = \{\overline{Q_1}, \dots, \overline{Q_n}\}$
 - ▶ $\overline{Q_1}$ is **outstanding** in $E \uplus H$
 - ▶ $E \uplus H$ is the first conflict with an **outstanding** assignment
 - ▶ $\text{level}_\Gamma(\overline{Q_1})$ is the current one (maximum on the trail)
 - ▶ Q_1 is the assertion literal in the first assertion clause
 $Q_1 \vee \dots \vee Q_n$
- ▶ Let $H' = \{\overline{Q_2}, \dots, \overline{Q_n}\}$
- ▶ **Backjump** to $\text{level}_\Gamma(E \uplus H')$ adding $E \uplus H' \vdash Q_1$

The CDCL example in CDSAT

$$S = \{\bar{A} \vee B, \bar{A} \vee C \vee E, \bar{B} \vee D, \bar{C} \vee \bar{D}, A \vee \bar{B} \vee E, B \vee \bar{C}, F \vee \bar{E}\}$$

subset of input

1. **Decide** adds $? \bar{F}$ to trail Γ opening level n
2. **Deduce** adds $J \vdash \bar{E}$ with $J = \{F \vee \bar{E}, \bar{F}\}$ to level n
since $\{F \vee \bar{E}, \bar{F}\} \vdash_{\text{Bool}} \bar{E}$
3. Two more **Decide** create levels $n + 1$ and $n + 2$
4. Another **Decide** adds $?A$ opening level $n + 3$
5. **Deduce** adds to level $n + 3$
 $H \vdash B$ with $H = \{\bar{A} \vee B, A\}$
 $I \vdash C$ with $I = \{\bar{A} \vee C \vee E, \bar{E}, A\}$
 $K \vdash D$ with $K = \{\bar{B} \vee D, B\}$

The CDCL example in CDSAT

6. $\{\overline{C} \vee \overline{D}, C\} \vdash_{\text{Bool}} \overline{D}$ but $\kappa \vdash D \in \Gamma$
Conflict: $E_0 = \{\overline{C} \vee \overline{D}, C, D\}$
/* $\overline{C} \vee \overline{D}$ conflict clause, not assertion clause */
7. E_0 contains literals C and D of max level $(n+3)$
Resolve: $E_1 = \{\overline{C} \vee \overline{D}, C, \overline{B} \vee D, B\}$
/* $\overline{C} \vee \overline{D}$ and $\overline{B} \vee D$ yield $\overline{B} \vee \overline{C}$ (not assertion clause) */
8. E_1 contains literals C and B of max level $(n+3)$
Resolve: $E_2 = \{\overline{C} \vee \overline{D}, \overline{A} \vee C \vee E, \overline{E}, A, \overline{B} \vee D, B\}$
/* $\overline{B} \vee \overline{C}$ and $\overline{A} \vee C \vee E$ yield $\overline{B} \vee \overline{A} \vee E$ (not assertion clause) */
9. E_2 contains literals A and B of max level $(n+3)$
Resolve: $E_3 = \{\overline{C} \vee \overline{D}, \overline{A} \vee C \vee E, \overline{E}, A, \overline{B} \vee D, \overline{A} \vee B\}$
/* $\overline{B} \vee \overline{A} \vee E$ and $\overline{A} \vee B$ yield $\overline{A} \vee E$ (assertion clause) */

The CDCL example in CDSAT with Backjump

$$E_3 = \{\overline{C} \vee \overline{D}, \overline{A} \vee C \vee E, \overline{E}, \textcolor{red}{A}, \overline{B} \vee D, \overline{A} \vee B\}$$

$\textcolor{red}{A}$ has level $n+3$ and is **outstanding** in E_3 ,

$\overline{E}$ has level n , and the rest has level 0

10. **Backjump** to level n adding $_G\vdash(\overline{A})$ with

$$G = \{\overline{C} \vee \overline{D}, \overline{A} \vee C \vee E, \overline{E}, \overline{B} \vee D, \overline{A} \vee B\}$$

11. **Deduce** adds to level n :

$$_N\vdash\overline{B} \text{ with } N = \{A \vee \overline{B} \vee E, \overline{A}, \overline{E}\}$$

$$_P\vdash\overline{C} \text{ with } P = \{B \vee \overline{C}, \overline{B}\}$$

Γ contains $\{\overline{E}, \overline{A}, \overline{B}, \overline{C}\}$ model of S

However, the assertion clause $\overline{A} \vee E$ is not learned

The CDSAT transition system

For the Boolean case (only Boolean assignments):

- ▶ Basic transition system:
 - ▶ Trail rules: **Decide**, **Deduce**, **Fail**, **ConflictSolve**
 - ▶ Conflict state rules: **Resolve**, **Backjump**
- ▶ Standard transition system:
 - ▶ Trail rules: **Decide**, **Deduce**, **Fail**, **ConflictSolve**
 - ▶ Conflict state rules: **Resolve**, **LearnBackjump**

LearnBackjump generalizes **Backjump** by learning any clause $L_1 \vee \dots \vee L_k$ which is the flip of a cube $\{\bar{L}_1, \dots, \bar{L}_k\}$ in the conflict

Conflict state rule LearnBackjump

- ▶ Conflict state: $\langle \Gamma; E \uplus H \rangle$ where $H = \{\bar{L}_1, \dots, \bar{L}_k\}$
- ▶ Let $L = L_1 \vee \dots \vee L_k$: **clausal form** of H
- ▶ Transition rule **LearnBackjump** solves the conflict by jumping back to a level m such that $\text{level}_\Gamma(E) \leq m < \text{level}_\Gamma(H)$ and adding ${}_E \vdash L$, provided $L \notin \Gamma$, $\bar{L} \notin \Gamma$:

$$\text{LearnBackjump: } \langle \Gamma; E \uplus H \rangle \Longrightarrow \Gamma^{\leq m}, {}_E \vdash L$$

- ▶ $(H_0 \cup E \uplus H) \models \perp$ implies $(H_0 \cup E) \models L$: **justified assignment**
 ${}_E \vdash L$ is **sound** and L is a **learned lemma**: every model must satisfy L as long as $H_0 \cup E$ is on the trail

LearnBackjump generalizes Backjump

- ▶ Given conflict state $\langle \Gamma; E \uplus H \rangle$, if
 - ▶ H is a singleton $\{\bar{L}\}$ so that the **clausal form** of H is simply L
 - ▶ $\bar{L}$ is **outstanding** in $E \uplus \{\bar{L}\}$: $\text{level}_\Gamma(\bar{L}) > \text{level}_\Gamma(E)$
 - ▶ The destination level m is $\text{level}_\Gamma(E)$
- ▶ Then **LearnBackjump** simulates **Backjump**

First assertion clause heuristic with LearnBackjump

Apply **Resolve** until **conflict** has the form $E \uplus H$ where

- ▶ $H = \{\overline{Q_1}, \dots, \overline{Q_n}\}$
- ▶ The clausal form of H is the first assertion clause $C = Q_1 \vee \dots \vee Q_n$ with assertion literal Q_1
- ▶ $\overline{Q_1}$ is **outstanding** in $E \uplus H$
- ▶ $E \uplus H$ is the first conflict with an **outstanding** literal
- ▶ $\text{level}_\Gamma(\overline{Q_1})$ is the current one (maximum on the trail)

Let $H' = \{\overline{Q_2}, \dots, \overline{Q_n}\}$

First assertion clause heuristic with LearnBackjump

- ▶ **LearnBackjump** to level $m = \text{level}_\Gamma(E \uplus H')$ adding $E \vdash C$
- ▶ The condition $\text{level}_\Gamma(E) \leq m < \text{level}_\Gamma(H)$ is met:
 - ▶ $m < \text{level}_\Gamma(H)$ as $\text{level}_\Gamma(H) = \text{level}_\Gamma(\overline{Q_1})$ which is the current one (maximum on the trail)
 - ▶ $m \geq \text{level}_\Gamma(E)$ as $E \uplus H'$ is a superset of E
- ▶ Apply **Deduce** to add $\{C\} \uplus H' \vdash Q_1$ supported by unit propagation inference $\{C\} \uplus H' \vdash_{\text{Bool}} Q_1$

LearnBackjump allows other heuristics:

e.g., **learn and restart** when $m = \text{level}_\Gamma(E) = 0$

The CDCL example in CDSAT with LearnBackjump

$$E_3 = \{\overline{C} \vee \overline{D}, \overline{A} \vee C \vee E, \overline{E}, A, \overline{B} \vee D, \overline{A} \vee B\}$$

A has level $n + 3$ and is **outstanding** in E_3 ,

$\overline{E}$ has level n , and the rest has level 0

10. **LearnBackjump** to level n adding $G \vdash (\overline{A} \vee E)$ with

$$G = \{\overline{C} \vee \overline{D}, \overline{A} \vee C \vee E, \overline{B} \vee D, \overline{A} \vee B\}$$

11. **Deduce** adds $M \vdash \overline{A}$ to level n with $M = \{\overline{A} \vee E, \overline{E}\}$

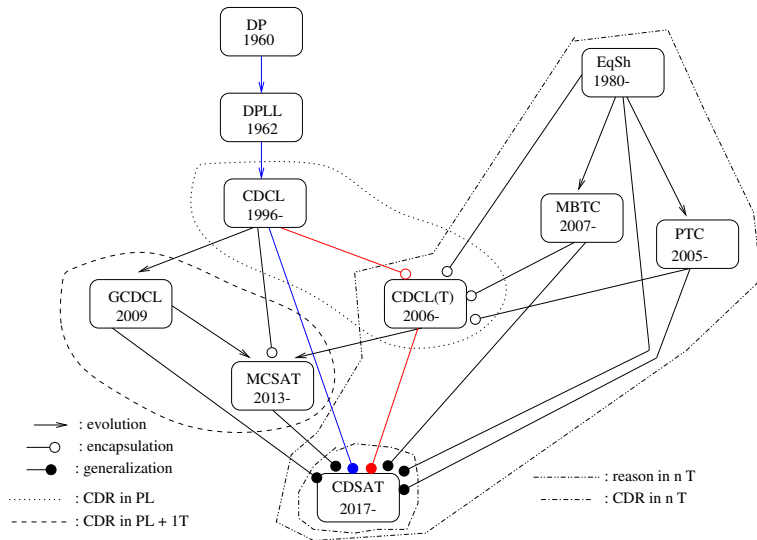
12. **Deduce** adds to level n :

$$N \vdash \overline{B} \text{ with } N = \{A \vee \overline{B} \vee E, \overline{A}, \overline{E}\}$$

$$P \vdash \overline{C} \text{ with } P = \{B \vee \overline{C}, \overline{B}\}$$

Γ contains $\{\overline{E}, \overline{A}, \overline{B}, \overline{C}\}$ model of S

The big picture: from SAT to SMT



CDCL($\mathcal{T}$): from SAT to SMT for one theory

DPLL($\mathcal{T}$) later renamed CDCL($\mathcal{T}$) for $\mathcal{T}$ a single theory
[Nieuwenhuis, Oliveras, Tinelli: JACM 2006]

- ▶ CDCL + decision procedure for the $\mathcal{T}$ -satisfiability of a set of $\mathcal{T}$ -literals
- ▶ CDCL works on a propositional abstraction:
input $\mathcal{T}$ -atoms replaced by propositional variables

CDCL($\mathcal{T}$): from SAT to SMT for one theory

Let $\{L_1, \dots, L_n\} \subseteq \Gamma$ and $C = \bar{L}_1 \vee \dots \vee \bar{L}_n$

The $\mathcal{T}$ -sat procedure contributes only:

- ▶ $\mathcal{T}$ -conflict detection:
if $\{L_1, \dots, L_n\} \models_{\mathcal{T}} \perp$
 C is a conflict clause
- ▶ $\mathcal{T}$ -propagation:
if $\{L_1, \dots, L_n\} \models_{\mathcal{T}} L$ and $L \notin \Gamma$ and $\bar{L} \notin \Gamma$
add L to Γ with justification $C \vee L$:
the atom of L **must be** an input atom (i.e., **not new**)
already mapped to a propositional variable

CDCL($\mathcal{T}$): from SAT to SMT for one theory

- ▶ $\mathcal{T}$ -sat procedure integrated as a **black-box** that only raises a flag if it detects an inconsistency in the propositional model built by CDCL ignoring the theory:
 - ▶ **$\mathcal{T}$ -conflict**: $\{L_1, \dots, L_n\} \models_{\mathcal{T}} \perp$
 $\bar{L}_1 \vee \dots \vee \bar{L}_n$ is $\mathcal{T}$ -valid
 - ▶ **$\mathcal{T}$ -propagation**: $\{L_1, \dots, L_n, \bar{L}\} \models_{\mathcal{T}} \perp$
 $\bar{L}_1 \vee \dots \vee \bar{L}_n \vee L$ is $\mathcal{T}$ -valid

Never deduce anything that excludes a $\mathcal{T}$ -model but is not a $\mathcal{T}$ -valid clause or a logical consequence of the input by propositional resolution

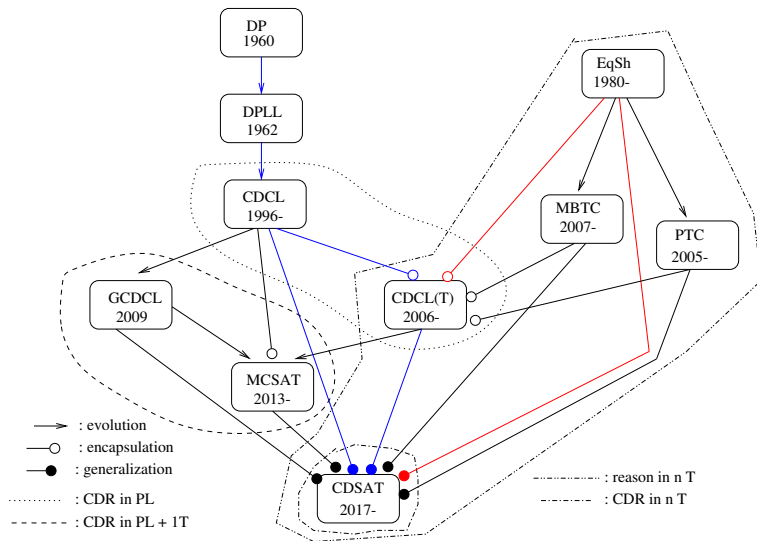
- ▶ Model search, trail, conflict explanation, conflict-driven reasoning remain propositional

CDSAT generalizes CDCL($\mathcal{T}$)

- ▶ Consider a theory union whose members are **Bool** and $\mathcal{T}$
- ▶ Theory modules:
 - ▶ **Bool**-module
 - ▶ **Black-box** $\mathcal{T}$ -module:
 - ▶ Only one inference rule: $L_1, \dots, L_m \vdash \perp$
 - ▶ That invokes the $\mathcal{T}$ -procedure to detect that a set of literals is $\mathcal{T}$ -unsatisfiable

CDSAT can use a **black-box** $\mathcal{T}$ -module
whenever a theory $\mathcal{T}$ is not involved in conflict-driven reasoning

The big picture: theory combination



Classical approach to theory combination: equality sharing

Equality sharing mostly known as **Nelson-Oppen method**

[Nelson, Oppen: ACM TOPLAS 1979], [Nelson: ATP After 25 Years 1983],
[Tinelli, Harandi: FroCoS 1996]

- ▶ $\mathcal{T} = \bigcup_{k=1}^n \mathcal{T}_k$: **disjoint** theories (share $\simeq$ and sorts)
- ▶ Decision procedures for $\mathcal{T}_k$ -satisfiability of set of $\mathcal{T}_k$ -literals
- ▶ **Stably infinite**: $\mathcal{T}_k$ -model with infinite cardinality
- ▶ Get decision procedure for $\mathcal{T}$ -satisfiability of set of $\mathcal{T}$ -literals
- ▶ Combination of decision procedures as **black-boxes**
- ▶ By **disjointness**, agreement is needed only on:
 - ▶ Cardinalities of shared sorts: by stable infiniteness
 - ▶ Interpretation of equality

Equality sharing: separation

- ▶ Input set S : $\mathcal{T}$ -literals mix symbols from the $\mathcal{T}_k$'s signatures
- ▶ **Separate** S into sets S_k of $\mathcal{T}_k$ -literals ($1 \leq k \leq n$)
- ▶ The sets S_k have only $\simeq$ and variables in common

Example of separation

- ▶ $\{x \leq y, y \leq (x + g(x)), P(h(x) - h(y)), \neg P(0), g(x) \simeq 0\}$
- ▶ Theory combination: **LIA** and **EUF** sharing sorts Z and prop
- ▶ **EUF**-procedure: congruence closure algorithm
[Nelson, Oppen: JACM 1980] [Downey, Sethi, Tarjan: JACM 1980]
- ▶ Standard elimination of predicate symbols other than $\simeq$:
 $S = \{x \leq y, y \leq (x + g(x)),$
 $f_p(h(x) - h(y)) \simeq \bullet, f_p(0) \not\simeq \bullet, g(x) \simeq 0\}$

Example of separation

$$S = \{x \leq y, y \leq (x + g(x)), \\ f_p(h(x) - h(y)) \simeq \bullet, f_p(0) \not\simeq \bullet, g(x) \simeq 0\}$$

Separation:

- ▶ $S_{LIA} = \{x \leq y, y \leq x + w_1, w_3 - w_4 \simeq w_2, w_5 \simeq 0, w_1 \simeq 0\}$
- ▶ $S_{EUF} =$
 $\{g(x) \simeq w_1, f_p(w_2) \simeq \bullet, h(x) \simeq w_3, h(y) \simeq w_4, f_p(w_5) \not\simeq \bullet\}$
- ▶ Set of shared variables: $\{x, y, w_1, w_2, w_3, w_4, w_5\}$

Equality sharing: the reduction

- ▶ Reduce the $\mathcal{T}$ -sat problem to $\mathcal{T}_k$ -sat problems
- ▶ S is $\mathcal{T}$ -sat iff $\bigcup_{k=1}^n S_k$ is $\mathcal{T}$ -sat
- ▶ **Arrangement** α : represents a **partition** of the set of shared variables
- ▶ Conjunction $\alpha = \alpha^+ \wedge \alpha^-$ where
 - ▶ $(u \simeq v) \in \alpha^+$ if u and v in the same class
 - ▶ $(u \not\simeq v) \in \alpha^-$ otherwise
- ▶ **Combination theorem**:
 $\bigcup_{k=1}^n S_k$ is $\mathcal{T}$ -sat iff $\exists \alpha$ s.t. $S_k \wedge \alpha$ is $\mathcal{T}_k$ -sat ($1 \leq k \leq n$)

A theory $\mathcal{T}_k$ is **convex** if for all conjunctions H of $\mathcal{T}_k$ -literals $H \models_{\mathcal{T}_k} \bigvee_{i=1}^n u_i \simeq v_i$ implies $\exists j, 1 \leq j \leq n, H \models_{\mathcal{T}_k} u_j \simeq v_j$.

Meaning: $H \models_{\mathcal{T}_k} \bigvee_{i=1}^n u_i \simeq v_i$ means that for all $\mathcal{T}_k$ -models $\mathcal{M}$ of H there exists an $i, 1 \leq i \leq n$, such that $\mathcal{M} \models_{\mathcal{T}_k} u_i \simeq v_i$, e.g.:

$\mathcal{M}_1 \models_{\mathcal{T}_k} u_1 \simeq v_1$,

$\mathcal{M}_2 \models_{\mathcal{T}_k} u_2 \simeq v_2$,

$\mathcal{M}_3 \models_{\mathcal{T}_k} u_1 \simeq v_1$, and so on.

Convexity means that there exists a $j, 1 \leq j \leq n$, such that for all $\mathcal{T}_k$ -models $\mathcal{M}$ of H it is $\mathcal{M} \models_{\mathcal{T}_k} u_j \simeq v_j$.

Equality sharing: build arrangement (convex theories)

$\mathcal{E}$: shared set of equalities between shared variables

- ▶ $\mathcal{E}_0 = \emptyset$
- ▶ $\mathcal{E}_i = \mathcal{E}_{i-1} \cup \{u \simeq v\}$ if a $\mathcal{T}_k$ -procedure deduces $u \simeq v$ from $S_k \cup \mathcal{E}_{i-1}$
(every $\mathcal{T}_k$ -procedure must be able to deduce **all** entailed equalities)
- ▶ If a $\mathcal{T}_k$ -sat procedure deduces $\perp$ from $S_k \cup \mathcal{E}_i$ for some i :
Return **unsat** (S is **$\mathcal{T}$ -unsat**)
- ▶ If neither $\perp$ nor additional equalities arise
(i.e., $\mathcal{E}_q = \mathcal{E}_{q-1}$ for some q): let $\alpha^+ = \mathcal{E}_q$
let α^- contain $u \neq v$ for all (u, v) not in $\mathcal{E}_q$
Return **sat** (S is **$\mathcal{T}$ -sat**)

Equality sharing: build arrangement (non-convex theories)

- ▶ $\mathcal{T}_k$ **not convex**: $\mathcal{T}_k$ -procedure deduces $\bigvee_{j=1}^m u_j \simeq v_j$
(every $\mathcal{T}_k$ -procedure must be capable of deducing **all** entailed disjunctions of equalities)
- ▶ $\bigvee_{j=1}^m u_j \simeq v_j$ is subject to splitting
- ▶ The $\mathcal{T}$ -procedure (implementing the equality sharing method) calls itself recursively on each subproblem obtained by adding $u_j \simeq v_j$ to the current $\mathcal{E}_i$
- ▶ If a branch returns **sat** then return **sat** (S is **$\mathcal{T}$ -sat**)
- ▶ If all branches return **unsat** then return **unsat** (S is **$\mathcal{T}$ -unsat**)

Equality sharing: example

- ▶ $\{x \leq y, y \leq (x + g(x)), P(h(x) - h(y)), \neg P(0), g(x) \simeq 0\}$
- ▶ Theory combination: **LIA** and **EUF** sharing sorts Z and prop
- ▶ **EUF**-procedure: congruence closure algorithm
[Nelson, Oppen: JACM 1980] [Downey, Sethi, Tarjan: JACM 1980]
- ▶ Standard elimination of predicate symbols other than $\simeq$:
 $S = \{x \leq y, y \leq (x + g(x)),$
 $f_p(h(x) - h(y)) \simeq \bullet, f_p(0) \not\simeq \bullet, g(x) \simeq 0\}$

Equality sharing: example

$$S = \{x \leq y, y \leq (x + g(x)), \\ f_p(h(x) - h(y)) \simeq \bullet, f_p(0) \not\simeq \bullet, g(x) \simeq 0\}$$

Separation:

- ▶ $S_{LIA} = \{x \leq y, y \leq x + w_1, w_3 - w_4 \simeq w_2, w_5 \simeq 0, w_1 \simeq 0\}$
- ▶ $S_{EUF} =$
 $\{g(x) \simeq w_1, f_p(w_2) \simeq \bullet, h(x) \simeq w_3, h(y) \simeq w_4, f_p(w_5) \not\simeq \bullet\}$
- ▶ Set of shared variables: $\{x, y, w_1, w_2, w_3, w_4, w_5\}$

Equality sharing: example

$$S_{LIA} = \{x \leq y, y \leq x + w_1, w_3 - w_4 \simeq w_2, w_5 \simeq 0, w_1 \simeq 0\}$$

$$S_{EUF} =$$

$$\{g(x) \simeq w_1, f_p(w_2) \simeq \bullet, h(x) \simeq w_3, h(y) \simeq w_4, f_p(w_5) \not\simeq \bullet\}$$

$$\mathcal{E}_0 = \emptyset$$

1. **LIA**-procedure: from $w_5 \simeq 0$ and $w_1 \simeq 0$ get $w_1 \simeq w_5$
 $\mathcal{E}_1 = \{w_1 \simeq w_5\}$
2. **LIA**-procedure: from $y \leq x + w_1$ and $w_1 \simeq 0$ get $y \leq x$
3. **LIA**-procedure: from $x \leq y$ and $y \leq x$ get $x \simeq y$
 $\mathcal{E}_2 = \{w_1 \simeq w_5, x \simeq y\}$

Equality sharing: example

$$S_{LIA} = \{x \leq y, y \leq x + w_1, w_3 - w_4 \simeq w_2, w_5 \simeq 0, w_1 \simeq 0\}$$

$$S_{EUF} =$$

$$\{g(x) \simeq w_1, f_p(w_2) \simeq \bullet, h(x) \simeq w_3, h(y) \simeq w_4, f_p(w_5) \not\simeq \bullet\}$$

$$\mathcal{E}_2 = \{w_1 \simeq w_5, x \simeq y\}$$

4. **EUF**-procedure: from $x \simeq y$ get $h(x) \simeq h(y)$
5. **EUF**-procedure: from $h(x) \simeq h(y)$, $h(x) \simeq w_3$, and $h(y) \simeq w_4$
get $w_3 \simeq w_4$
 $\mathcal{E}_3 = \{w_1 \simeq w_5, x \simeq y, w_3 \simeq w_4\}$
6. **LIA**-procedure: from $w_3 \simeq w_4$ get $w_3 - w_4 \simeq 0$
7. **LIA**-procedure: from $w_3 - w_4 \simeq 0$ and $w_3 - w_4 \simeq w_2$
get $w_2 \simeq 0$

Equality sharing: example

$$S_{LIA} = \{x \leq y, y \leq x + w_1, w_3 - w_4 \simeq w_2, w_5 \simeq 0, w_1 \simeq 0\}$$

$$S_{EUF} =$$

$$\{g(x) \simeq w_1, f_p(w_2) \simeq \bullet, h(x) \simeq w_3, h(y) \simeq w_4, f_p(w_5) \not\simeq \bullet\}$$

$$\mathcal{E}_3 = \{w_1 \simeq w_5, x \simeq y, w_3 \simeq w_4\}$$

8. **LIA**-procedure: from $w_2 \simeq 0$ and $w_1 \simeq 0$ get $w_2 \simeq w_1$

$$\mathcal{E}_4 = \{w_1 \simeq w_5, x \simeq y, w_3 \simeq w_4, w_2 \simeq w_1\}$$

9. **LIA**-procedure: from $w_2 \simeq 0$ and $w_5 \simeq 0$ get $w_2 \simeq w_5$

$$\mathcal{E}_5 = \{w_1 \simeq w_5, x \simeq y, w_3 \simeq w_4, w_2 \simeq w_1, w_2 \simeq w_5\}$$

10. **EUF**-procedure: from $w_2 \simeq w_5$ get $f_p(w_2) \simeq f_p(w_5)$

Equality sharing: example

$$S_{\text{LIA}} = \{x \leq y, y \leq x + w_1, w_3 - w_4 \simeq w_2, w_5 \simeq 0, w_1 \simeq 0\}$$

$$S_{\text{EUF}} =$$

$$\{g(x) \simeq w_1, f_p(w_2) \simeq \bullet, h(x) \simeq w_3, h(y) \simeq w_4, f_p(w_5) \not\simeq \bullet\}$$

$$\mathcal{E}_5 = \{w_1 \simeq w_5, x \simeq y, w_3 \simeq w_4, w_2 \simeq w_1, w_2 \simeq w_5\}$$

11. EUF-procedure: from $f_p(w_2) \simeq f_p(w_5)$ and $f_p(w_2) \simeq \bullet$
get $f_p(w_5) \simeq \bullet$
12. EUF-procedure: from $f_p(w_5) \simeq \bullet$ and $f_p(w_5) \not\simeq \bullet$
get $\perp$ and return **unsat**

Equality sharing is not conflict-driven

- ▶ Combining theories by combining procedures
- ▶ $\mathcal{T}_k$ -procedures combined as **black-boxes** sharing only equalities btw shared variables
- ▶ Generation of all entailed (disjunctions of) equalities resembles **saturation** (equality sharing can be emulated by **superposition** for theories that satisfy certain hypotheses)

CDCL($\mathcal{T}$) and equality sharing

CDCL($\mathcal{T}$) where $\mathcal{T}$ -procedure is equality sharing combination

[Barrett et al.: LPAR 2006], [Krstić, Goel: FroCoS 2007]

- ▶ $\mathcal{T}$ -procedure sends (propositional abstraction of)
 $\bigvee_{j=1}^m u_j \simeq v_j$ to CDCL
- ▶ Reasoning about disjunction is entrusted to CDCL
- ▶ Case $u_j \simeq v_j$ is considered when CDCL puts it on the trail
(instance of splitting on demand)
- ▶ Sole new (i.e., non-input) literals in CDCL($\mathcal{T}$):
(propositional abstractions of) equalities btw shared variables
- ▶ Model search, trail, conflict explanation, conflict-driven reasoning remain propositional

CDSAT can emulate equality sharing

Towards how CDSAT can emulate equality sharing:

1. What CDSAT does for separation
2. More about CDSAT theory modules: [equality inferences](#)
3. The trail remains Boolean (equalities are Boolean terms) but is **shared** by more than one theory module

CDSAT: foreign terms as variables

$\triangleleft$: proper subterm ordering

$top(t)$: top symbol of term t

- ▶ A symbol that is in $\mathcal{T}_j$'s signature but not in $\mathcal{T}_k$'s signature ($k \neq j$) is **foreign** to $\mathcal{T}_k$ or $\mathcal{T}_k$ -**foreign**
- ▶ A term t is $\mathcal{T}_k$ -**foreign** if $top(t)$ is
- ▶ Each $\mathcal{T}_k$ sees as a variable a $\triangleleft$ -maximal occurrence of a $\mathcal{T}_k$ -**foreign** term

Why $\triangleleft$ -maximal occurrences?

Example:

- ▶ Theory $\mathcal{T}_k$
- ▶ Symbol f is in $\mathcal{T}_k$'s signature and g is not
- ▶ Symbols g is $\mathcal{T}_k$ -foreign and f is not
- ▶ Term $f(\mathbf{g}(x), \mathbf{g}(g(y)))$
- ▶ $\mathcal{T}_k$ sees $\mathbf{g}(x)$ and $\mathbf{g}(g(y))$ as variables, not $g(y)$
- ▶ $g(y)$ is not visible once $\mathbf{g}(g(y))$ is seen as a variable

- ▶ Given: an assignment H
- ▶ The set of **shared terms** $\mathcal{V}_{\text{sh}}(H)$ consists of:
 - ▶ Assigned terms
 - ▶ Foreign terms (seen as variables by at least one theory)
 - ▶ Shared variables including both
 - ▶ Original variables that are shared and
 - ▶ Foreign terms that are shared variables because they are foreign to more than one theory

The separation example done by CDSAT

- ▶ $H = \{x \leq y, y \leq (x + g(x)), p(h(x) - h(y)), \overline{p(0)}, g(x) \simeq 0\}$
- ▶ Theory union: **LIA** $\cup$ **EUF** sharing sorts Z and prop
- ▶ **EUF**-module: abstraction of the congruence closure algorithm
- ▶ **EUF**: $\leq, +, -, \text{ and } 0$ are foreign symbols
 $x \leq y$ and $y \leq (x + g(x))$ seen as variables of sort prop
 $h(x) - h(y)$ and 0 seen as variables of sort Z
- ▶ **LIA**: $g, p,$ and h are foreign symbols
 $g(x)$ seen as a variable of sort Z
 $p(h(x) - h(y))$ and $p(0)$ seen as variables of sort prop

The separation example done by CDSAT

- ▶ **Shared terms:** assigned terms + foreign terms + shared vars
- ▶ Assigned terms:
 $\{x \leq y, y \leq (x + g(x)), p(h(x) - h(y)), p(0), g(x) \simeq 0\}$
- ▶ Foreign terms: **EU**F-foreign + **LIA**-foreign
EUF-foreign: $\{x \leq y, y \leq (x + g(x)), h(x) - h(y), 0\}$
LIA-foreign: $\{g(x), p(h(x) - h(y)), p(0)\}$
- ▶ Shared vars: $\{x\}$: **LIA** sees $\{x, y\}$; **EU**F sees $\{x\}$ as all occurrences of y are below **EU**F-foreign terms
- ▶ **Shared terms:** $\mathcal{V}_{\text{sh}}(H) = \{x \leq y, y \leq (x + g(x)), p(h(x) - h(y)), p(0), g(x) \simeq 0, 0, h(x) - h(y), g(x), p(0), p(h(x) - h(y)), x\}$

Separation in Equality Sharing vs. CDSAT: comparison

- ▶ Shared variables for equality sharing: $\{x, y, w_1, w_2, w_3, w_4, w_5\}$
 w_1 stands for $g(x)$, w_2 stands for $h(x) - h(y)$, w_3 stands for $h(x)$
 w_4 stands for $h(y)$, w_5 stands for 0
- ▶ **Shared terms** for CDSAT: $\mathcal{V}_{\text{sh}}(H) = \{x \leq y, y \leq (x + g(x)), p(h(x) - h(y)), p(0), g(x) \simeq 0, 0, h(x) - h(y), g(x), p(0), p(h(x) - h(y)), x\}$
- ▶ $h(x)$ and $h(y)$ not in $\mathcal{V}_{\text{sh}}(S)$: they are below **EU**F-foreign term $h(x) - h(y)$ and **LI**A-foreign term $p(h(x) - h(y))$
- ▶ Assigned terms are shared in CDSAT as they sit on a **shared** trail, not in equality sharing where each **black-box** procedure has its own literals

CDSAT modules: equality inference rules

All CDSAT theory modules include **equality inference rules**:

- ▶ Reflexivity: $\vdash t \simeq t$
- ▶ Symmetry: $t \simeq s \vdash s \simeq t$
- ▶ Transitivity: $t \simeq s, s \simeq u \vdash t \simeq u$
- ▶ Same value: $t \leftarrow c, s \leftarrow c \vdash t \simeq s$
- ▶ Different values: $t \leftarrow c, s \leftarrow q \vdash t \not\simeq s$
- ▶ All modules know that equality is an **equivalence**
- ▶ Only the module for the theory of equality (**EUF**) knows that it is a **congruence**

CDSAT module for equality with uninterpreted functions

- ▶ $\Sigma_{\text{EUF}} = \langle S, F \rangle$
- ▶ $\text{prop} \in S$
- ▶ $\simeq_s \in F$ for all sorts $s \in S$
- ▶ EUF^+ may be
 - ▶ **Trivial** or
 - ▶ **Countable**: values used as labels of congruence classes, e.g.:
 $t_1 \leftarrow c, t_2 \leftarrow c, t_3 \leftarrow c_3, t_4 \leftarrow c_4, t_5 \leftarrow c_5$
shorter than
 $t_1 \simeq t_2, t_1 \not\simeq t_3, t_1 \not\simeq t_4, t_1 \not\simeq t_5, t_3 \not\simeq t_4, t_3 \not\simeq t_5, t_4 \not\simeq t_5$

CDSAT module for equality with uninterpreted functions

- ▶ Inference rules for congruence:
 - ▶ $(t_i \simeq u_i)_{i=1\dots m}, (f(t_1, \dots, t_m) \not\simeq f(u_1, \dots, u_m)) \vdash_{\text{EUF}} \perp$
 - ▶ $(t_i \simeq u_i)_{i=1\dots m} \vdash_{\text{EUF}} f(t_1, \dots, t_m) \simeq f(u_1, \dots, u_m)$
 - ▶ $(t_i \simeq u_i)_{i=1\dots m, i \neq j}, f(t_1, \dots, t_m) \not\simeq f(u_1, \dots, u_m) \vdash_{\text{EUF}} t_j \not\simeq u_j$
- ▶ $\text{basis}_{\text{EUF}}(X)$: all subterms of terms in X and all equalities between them

CDSAT can emulate equality sharing

- ▶ The **Deduce** transition rule of CDSAT covers $\mathcal{T}_k$ -propagation
- ▶ Each $\mathcal{T}_k$ module can place an **inference** $J \vdash_k L$ as justified assignment $J \vdash L$ on the **shared trail**, including:
 - ▶ Equality inferences deducing $u \simeq v$
 - ▶ **EU**F-module inferences deducing $u \simeq v$ by congruence
 - ▶ Other $\mathcal{T}_k$ -specific inferences deducing $u \simeq v$
 - ▶ $\mathcal{T}_k$ -specific inferences deducing $\bigvee_{j=1}^m u_j \simeq v_j$ in non-convex $\mathcal{T}_k$

CDSAT can emulate equality sharing

- ▶ The $\mathcal{T}_k$ modules cooperate to build an arrangement **publicly** on the **shared trail**
- ▶ The **Bool**-module can handle disjunctions, if any, by **decision** and **unit propagation**
- ▶ However, in CDSAT the arrangement
 1. Is about the set $\mathcal{V}_{\text{sh}}(H)$ of **shared terms**, not the set of shared variables of equality sharing
 2. Does **not** have to be **complete**: a **partial** arrangement of $\mathcal{V}_{\text{sh}}(H)$ suffices for completeness, if Γ is **model-describing**

The equality sharing example done by CDSAT

$$H = \{x \leq y, y \leq (x + g(x)), p(h(x) - h(y)), \overline{p(0)}, g(x) \simeq 0\}$$

1. **LIA**-module: $\{y \leq x + g(x), g(x) \simeq 0\} \vdash_{\text{LIA}} y \leq x$

Deduce: $J \vdash (y \leq x)$ (level 0)

with $J = \{y \leq x + g(x), g(x) \simeq 0\}$

2. **LIA**-module: $\{x \leq y, y \leq x\} \vdash_{\text{LIA}} x \simeq y$

Deduce: $H \vdash (x \simeq y)$ (level 0)

with $H = \{x \leq y, y \leq x\}$

3. **EUF**-module: $x \simeq y \vdash_{\text{EUf}} h(x) \simeq h(y)$

Deduce: $I \vdash (h(x) \simeq h(y))$ (level 0)

with $I = \{x \simeq y\}$

The equality sharing example done by CDSAT

4. **LIA**-module: $h(x) \simeq h(y) \vdash_{\text{LIA}} h(x) - h(y) \simeq 0$
Deduce: $K \vdash (h(x) - h(y) \simeq 0)$ (level 0)
with $K = \{h(x) \simeq h(y)\}$
5. **EUF**-module: $\{h(x) - h(y) \simeq 0\} \vdash_{\text{EUF}} p(h(x) - h(y)) \simeq p(0)$
Deduce: $N \vdash (p(h(x) - h(y)) \simeq p(0))$ (level 0)
with $N = \{h(x) - h(y) \simeq 0\}$
6. **EUF**-module: $\{(p(h(x) - h(y))) \leftarrow \text{true}, p(0) \leftarrow \text{false}\} \vdash_{\text{EUF}}$
 $p(h(x) - h(y)) \not\simeq p(0)$ (equality inference: also **LIA**-module)
but the trail contains $p(h(x) - h(y)) \simeq p(0)$: **conflict**
 $\{p(h(x) - h(y)), \overline{p(0)}, p(h(x) - h(y)) \simeq p(0)\}$ (level 0)
Fail returns **unsat**

CDSAT can handle the problem in a different way

- ▶ Each $\mathcal{T}_k$ module can place **decisions** on the **shared trail** by **Decide** transitions
- ▶ A $\mathcal{T}_k$ -inference $J \vdash_k L$ from $J \subseteq \Gamma$ leads to $\mathcal{T}_k$ -**conflict**
 $E = J \cup \{\bar{L}\}$ if $\bar{L} \in \Gamma$
- ▶ Solved by **Backjump** or **LearnBackjump**
- ▶ Deductions may detect and explain conflicts rather than being all in saturation mode as in equality sharing
- ▶ Interplay of decisions and deductions by different theory modules: **late propagation**

Late propagation

- ▶ Multiple theories and multiple theory modules **sharing** the trail
A justified assignment $H \vdash A$ may be added after assignments of higher level (decisions and possibly deductions by another module): **late propagation**
- ▶ It follows that the CDSAT trail is **not** a stack
- ▶ Equality sharing does not use a shared trail, because each procedure is a **black-box**
- ▶ The trail is a stack in
 - ▶ CDCL
 - ▶ $\text{CDCL}(\mathcal{T})$ where $\mathcal{T}$ is one theory
 - ▶ $\text{CDCL}(\mathcal{T})$ where $\mathcal{T}$ is an equality sharing combination

The equality sharing example done by CDSAT: variant

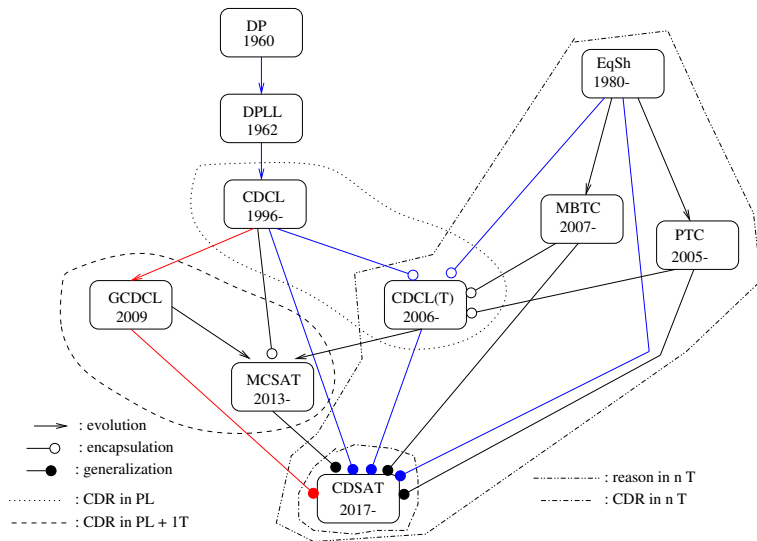
$$H = \{x \leq y, y \leq (x + g(x)), p(h(x) - h(y)), \overline{p(0)}, g(x) \simeq 0\}$$

1. EUF-module: **Decide** adds $?(x \neq y)$ (level 1)
2. LIA-module: $\{y \leq x + g(x), g(x) \simeq 0\} \vdash_{\text{LIA}} y \leq x$
Deduce: $J \vdash (y \leq x)$ (level 0)
with $J = \{y \leq x + g(x), g(x) \simeq 0\}$ /* late propagation */
3. LIA-module: $\{x \leq y, y \leq x\} \vdash_{\text{LIA}} x \simeq y$
but the trail contains $?(x \neq y)$
LIA-conflict: $E_0 = \{x \neq y, x \leq y, y \leq x\}$
The deduction of $x \simeq y$ detects a conflict

The equality sharing example done by CDSAT: variant

4. **LIA-conflict**: $E_0 = \{x \not\leq y, x \leq y, y \leq x\}$
 $?(x \not\leq y)$ has level 1, the rest has level 0
5. **Backjump** or **LearnBackjump**: back to level 0 adding
 $H \vdash (x \simeq y)$
with $H = \{x \leq y, y \leq x\}$
 $H \vdash (x \simeq y)$ added by a conflict solving transition
rather than a **Deduce** transition
the derivation may continue as before

The big picture: conflict-driven one-theory reasoning



Conflict-driven satisfiability procedures in arithmetic

- ▶ Input problem: set of literals in the theory signature
- ▶ Candidate model: theory model
- ▶ Assignments to **first-order** variables (e.g., $x \leftarrow 3$)
- ▶ **Propagation**: **evaluation** of arithmetic expressions
(e.g., $y \leftarrow 0 \vdash_{\text{LRA}} (y > 2) \leftarrow \text{false}$)
- ▶ **Explanation** of **theory-conflicts** by (expensive) **theory inferences**: used only **on demand** to respond to conflicts
- ▶ Learn lemmas that may contain **new** (non-input) atoms and may exclude first-order assignments, hence theory models
- ▶ Devise restrictions to ensure **termination**

Conflict-driven satisfiability procedures in arithmetic

Examples:

- ▶ Linear rational (or real) arithmetic (LRA):
Conflict Resolution [Korovin, Tsiskaridze, Voronkov: CP 2009]
GDPLL then GCDCL [McMillan, Kuehlmann, Sagiv: CAV 2009]
Natural Domain SMT [Cotton: FORMATS 2010]
- ▶ Linear integer arithmetic (LIA):
Cutting-to-the-chase procedure [Jovanovic, de Moura: CADE 2011, JAR 2013] [Bromberger, Sturm, Weidenbach: CADE 2015]
- ▶ Nonlinear real arithmetic (NRA):
NLSAT [Jovanovic, de Moura: IJCAR 2012]
k-SMT [Brauß, Korovin, Korovina, Müller: FroCoS 2019]

The language of LRA-literals

- ▶ Input: set S of LRA-literals
- ▶ LRA-term:
 - ▶ rational constant c
 - ▶ sum $c_1 \cdot x_1 + \dots + c_n \cdot x_n$
- ▶ LRA-literal: $t_1 \triangleleft t_2$ where $\triangleleft \in \{<, \leq\}$
- ▶ $\overline{(t_1 < t_2)}$ and $\overline{(t_1 \leq t_2)}$ replaced by $t_2 \leq t_1$ and $t_2 < t_1$
- ▶ $t_1 \simeq t_2$ rewritten as $t_1 \leq t_2$ and $t_2 \leq t_1$

The Fourier-Motzkin algorithm for LRA

- ▶ Decide the satisfiability of a set of LRA-literals
- ▶ Apply **variable elimination** until
 - ▶ Either a contradiction $0 \leq q$ with negative q is generated
Return **unsat**
 - ▶ Or all variables are eliminated
Returns **sat**
- ▶ **Termination guaranteed**: each round eliminates one variable
Finitely many variables

[Lassez, Maher: JAR 1992]

Variable elimination by linear combination in LRA

1. Select a variable x
2. If x appears only with positive, or negative, coefficients, remove all literals where x appears
3. Otherwise, compute **all linear combinations** of literals
 $t_1 + c_1 \cdot x \leq u_1$ and $t_2 - c_2 \cdot x \leq u_2$
generating the literal
 $c_2 \cdot t_1 + c_1 \cdot t_2 \leq c_2 \cdot u_1 + c_1 \cdot u_2$
(if a premise is strict, the result is also)

[Lassez, Maher: JAR 1992]

Variable elimination by transitive closure in LRA

3. Otherwise (recall $\triangleleft \in \{<, \leq\}$):
 - 3.1 Rearrange every literal where x has positive coefficient into upper bound $x \triangleleft t$
 - 3.2 Rearrange every literal where x has negative coefficient into lower bound $t \triangleleft x$
 - 3.3 Compute **all transitive closures** concatenating $u \leq x$ and $x \leq t$ to generate $u \leq t$
(if a premise is strict, the result is also)

The Fourier-Motzkin algorithm generates a doubly exponential number of literals in the worst case

[Schrijver: 1998], [Kroening, Strichman: 2008]

The level-saturation strategy in propositional logic

- ▶ Decide the satisfiability of a set of propositional clauses
- ▶ Apply **variable elimination** until
 - ▶ Either a contradiction $\{L, \bar{L}\}$ is generated
Return **unsat**
 - ▶ Or all variables are eliminated
Returns **sat**
- ▶ Termination guaranteed: each round eliminates one variable
Finitely many variables

[Chang, Lee: 1973]

Variable elimination by resolution in propositional logic

1. Select a propositional variable $/$
2. If $/$ appears only with positive, or negative, sign, remove all clauses where $/$ appears: **pure literal rule**
3. Otherwise:
 - ▶ Add all resolvents generated by resolving upon $/$
 - ▶ Remove all clauses where $/$ appears

The level-saturation strategy generates too many resolvents

An inference rule for LRA

- ▶ Extract from the Fourier-Motzkin algorithm an inference rule for **LRA** and use it in a **conflict-driven** manner like CDCL does with resolution
- ▶ **Fourier-Motzkin (FM) resolution**:
 $\{t_1 \triangleleft_1 x, x \triangleleft_2 t_2\} \vdash_{\text{LRA}} t_1 \triangleleft_3 t_2$
 $\triangleleft_1, \triangleleft_2, \triangleleft_3 \in \{<, \leq\}$
 $\triangleleft_3$ is $<$ if either $\triangleleft_1$ or $\triangleleft_2$ is $<$ and $\leq$ otherwise
 - ▶ Linear combination of constraints:
 $\{x + y < 0, -y + 2 < 0\} \vdash_{\text{LRA}} x + 2 < 0$
 - ▶ Transitive closure: $\{x < -y, -y < -2\} \vdash_{\text{LRA}} x < -2$

Conflict Resolution or GCDCL for LRA

It stands to the Fourier-Motzkin algorithm like CDCL stands to the level-saturation strategy for propositional logic

- ▶ Candidate LRA-model written as assignments to rational variables
- ▶ Propagation: evaluation of arithmetic expressions
- ▶ Explanation of LRA-conflicts by FM-resolution: used only on demand to respond to conflicts
- ▶ Learn lemmas that may contain new (non-input) atoms and may exclude LRA-models

Conflict-driven one-theory reasoning in CDSAT

In order to see how CDSAT covers conflict-driven procedures such as Conflict Resolution or GCDCL for [LRA](#):

1. First-order decisions $\gamma(t \leftarrow c)$: more about **acceptability**
2. Transition rule **Deduce** beyond propositional inferences (e.g., unit propagation) and deduction of equalities as in equality sharing
3. The CDSAT module for [LRA](#)
4. A CDSAT transition rule named **UndoClear** to solve conflicts due to first-order decisions

The CDSAT trail (beyond the Boolean case)

A trail Γ is a sequence of assignments $\Gamma = A_0, \dots, A_m$ where

- ▶ Each A_i is either a **decision** $?A$ or a **justified assignment** $H \vdash A$
- ▶ **Decision**: either **Boolean** $?L$ or **first-order** $?(t \leftarrow c)$ for c other than true or false
- ▶ **Justified assignment** $H \vdash A$:
 - ▶ Supported by an inference $H \vdash_k A$ in a theory module $\mathcal{I}_k$
 - ▶ Input assignment ($H = \emptyset$)
 - ▶ Due to conflict-solving transitions (**Backjump**, **LearnBackjump**)
 - ▶ All **Boolean** except for input **first-order** assignments
- ▶ Levels and the notion of conflict are unchanged except that also first-order assignments may appear in conflicts

More about acceptability of a decision: one theory case

Transition rule **Decide** adds to the trail a decision:

$$\text{Decide: } \Gamma \longrightarrow \Gamma, ?(u \leftarrow c)$$

provided $\mathcal{T}$ -assignment $u \leftarrow c$ is **acceptable** for $\mathcal{T}$ -module $\mathcal{I}$:

- ▶ Γ does not contain a $\mathcal{T}$ -assignment to term u
In the **Boolean** case: $(u \leftarrow \text{true}) \notin \Gamma$ and $(u \leftarrow \text{false}) \notin \Gamma$
- ▶ If $(u \leftarrow c)$ is a first-order $\mathcal{T}$ -assignment: for no $J \subseteq \Gamma$ there exists an $\mathcal{I}$ -inference $J \cup \{u \leftarrow c\} \vdash_{\mathcal{T}} L$ and $\bar{L} \in \Gamma$
Exclude a first-order decision that triggers an immediate conflict from which nothing can be learned

More on the CDSAT transition rule Deduce

- ▶ Propagation:
 - ▶ Boolean propagation: e.g., unit propagation
 - ▶ $\mathcal{T}_k$ -propagation: e.g., propagation of equalities when emulating equality sharing
- ▶ Preliminary conflict explanation with lemma learning:
 $\mathcal{T}_k$ -inferences explain a $\mathcal{T}_k$ -conflict
generating lemmas excluding $\mathcal{T}_k$ -assignments until the
 $\mathcal{T}_k$ -conflict can be detected as a Boolean conflict on the trail:
 $J \vdash_k L$ and $\bar{L} \in \Gamma$
unsatisfiable assignment $E = J \cup \{\bar{L}\}$

CDSAT module for linear rational arithmetic (LRA)

- ▶ Signature Σ_{LRA} :
 - ▶ Sorts: $S = \{\text{prop}, \mathbb{Q}\}$
 - ▶ Symbols: $\simeq_s$ for all $s \in S$ (s can be omitted)
 $1, +, <, \leq, q \cdot$ for all rational numbers $q \in \mathbb{Q}$
 - ▶ (e.g., $(-2 \cdot x + (-1 \cdot y)) < 0 \cdot 1$)
- ▶ Theory extension LRA^+ adds
 - ▶ Constants $\tilde{q}$ for all $q \in \mathbb{Q}$ ($\tilde{q}$ written q)
/* countable */
 - ▶ Axioms $\tilde{q} \simeq q \cdot 1$ for all $q \in \mathbb{Q}$
 - ▶ (e.g., $(-2 \cdot x + (-1 \cdot y)) < 0$ hence $-2x - y < 0$)

Summary of the first lecture

- ▶ Introduction and motivation
- ▶ Conflict-driven reasoning and CDSAT in a nutshell
- ▶ Propositional conflict-driven reasoning: CDSAT and CDCL
- ▶ Combination of theories: CDSAT and Nelson-Oppen
- ▶ Conflict-driven reasoning in one theory: LRA procedures

- ▶ Conflict-driven reasoning in one theory: LRA in CDSAT
- ▶ Conflict-driven reasoning in one theory from literals to clauses: CDSAT and MCSAT
- ▶ Conflict-driven reasoning in many theories: CDSAT
- ▶ CDSAT and arrays, maps, and vectors with abstract domain

CDSAT module for linear rational arithmetic (LRA)

Inference rules:

- ▶ **Evaluation:** $\{t_1 \leftarrow \tilde{q}_1, \dots, t_m \leftarrow \tilde{q}_m\} \vdash_{\text{LRA}} l \leftarrow b$
- ▶ **Positivization:** $\frac{}{t_1 \leq t_2} \vdash_{\text{LRA}} t_2 \leq t_1$
 $\frac{}{t_1 \leq t_2} \vdash_{\text{LRA}} t_2 < t_1$
- ▶ **Equality elimination:** $t_1 \simeq_Q t_2 \vdash_{\text{LRA}} \{t_1 \leq t_2, t_2 \leq t_1\}$

CDSAT module for linear rational arithmetic (LRA)

Inference rules:

- ▶ **FM-resolution:** $\{t_1 \triangleleft_1 x, x \triangleleft_2 t_2\} \vdash_{\text{LRA}} t_1 \triangleleft_3 t_2$
 $\triangleleft_1, \triangleleft_2, \triangleleft_3 \in \{<, \leq\}$
 $\triangleleft_3$ is $<$ if either $\triangleleft_1$ or $\triangleleft_2$ is $<$ and $\leq$ otherwise
- ▶ **Disequality elimination:**
 $\{t_1 \leq x, x \leq t_2, t_1 \simeq t_0, t_2 \simeq t_0, x \not\simeq t_0\} \vdash_{\text{LRA}} \perp$
detects an **LRA-conflict**: no value for variable x

Conflict state rule UndoClear

- ▶ Given conflict $E \uplus \{A\}$, assignment A is **outstanding** in $E \uplus \{A\}$ if $\text{level}_\Gamma(A) > \text{level}_\Gamma(E)$
- ▶ Conflict state: $\langle \Gamma; E \uplus \{A\} \rangle$ where A is an **outstanding first-order assignment** s.t $\text{level}_\Gamma(A) = m$
- ▶ Transition rule **UndoClear** solves the conflict by undoing A and removing all assignments of level greater than or equal to m :

$$\text{UndoClear: } \langle \Gamma; E \uplus \{A\} \rangle \Longrightarrow \Gamma^{\leq m-1}$$

- ▶ A is a **decision**: $m - 1 \geq 0$ implies $m \geq 1$, and first-order assignments in Γ are either **input** (level 0) or **decisions**

Conflict state rule UndoClear

- ▶ The removed assignments
 - ▶ Include $H \vdash L$ such that $A \in H$: if A goes, $H \vdash L$ must go
 - ▶ Are not necessarily the most recent ones, as there may have been **late propagations** that are more recent
- ▶ $\Gamma^{\leq m-1}$ is new (no loop) even if nothing is added:
 - ▶ A was **acceptable** when decided: no **conflict**
 - ▶ **Conflict** $E \uplus \{A\}$: some $H \vdash L$ such that $\text{level}_\Gamma(L) < m$ was deduced afterwards (**late propagation**)
 - ▶ $\Gamma^{\leq m-1}$ contains $H \vdash L$
- ▶ **UndoClear** fires after a **late propagation**

A simple example with UndoClear

$\{l_0: 2x + y \simeq 1, l_1: 2x + 2y \simeq 1\}$ subset of the input (level 0)

1. **Decide:** $?(x \leftarrow 0)$ (level 1) /* **acceptable** */
2. **Deduce:** $J \vdash (y \simeq 0)$ with $J = \{2x + y \simeq 1, 2x + 2y \simeq 1\}$ (level 0)
FM-resolution: $\{2x + y \simeq 1, 2x + 2y \simeq 1\} \vdash_{\text{LRA}} y \simeq 0$ ($l_1 - l_0$)
/* **late propagation** */
3. **Evaluation:** $\{x \leftarrow 0, y \simeq 0\} \vdash_{\text{LRA}} 2x + y \not\simeq 1$ detects
LRA-conflict $E = \{x \leftarrow 0, y \simeq 0, 2x + y \simeq 1\}$
UndoClear: undo $?(x \leftarrow 0)$ (**outstanding** in E) back to level 0
4. **Decide:** $?(x \leftarrow 1/2)$ (level 1)
/* only acceptable value for x : **forced decision** */

Forced decision

- ▶ If a first-order assignment $t \leftarrow c$ follows from the trail as c is the only **acceptable** value for t for a theory module, $t \leftarrow c$ is a **forced decision** for that module
- ▶ This is a reason why theory module inferences only infer **Boolean assignments**

Forced decision: examples

- ▶ $\{u \simeq t, t \leftarrow c\} \subseteq \Gamma$:
 $u \leftarrow c$ is a **forced decision** for the **EU**F-module
- ▶ $\{u \leq t, t \leq u, t \leftarrow c\} \subseteq \Gamma$
where $\leq$ is the ordering on the rationals:
 $u \leftarrow c$ is a **forced decision** for the **LRA**-module

How about termination of conflict-driven FM-resolution?

FM-resolutions **alternating on distinct variables** may diverge:

$$\begin{array}{llll} l_0 : & -2 \cdot x - y < 0 & & \\ l_1 : & x + y < 0 & & \\ l_2 : & x < -1 & & \\ l_3 : & -y < -2 & (l_0 + 2l_2) & \text{elim } x \\ l_4 : & x < -2 & (l_1 + l_3) & \text{elim } y \\ l_5 : & -y < -4 & (l_0 + 2l_4) & \text{elim } x \\ l_6 : & x < -4 & (l_1 + l_5) & \text{elim } y \\ l_7 : & -y < -8 & (l_0 + 2l_6) & \text{elim } x \\ \dots & \dots & \dots & \dots \end{array}$$

also when FM-resolution is applied only to respond to **conflicts**

The non-terminating LRA example in CDSAT

$l_0: -2 \cdot x - y < 0, l_1: x + y < 0, l_2: x < -1$ (level 0)

1. Decide: $?(y \leftarrow 0)$ (level 1) /* acceptable */
LRA-conflict: $\{-2 \cdot x - y < 0, x < -1, y \leftarrow 0\}$
2. Explained by $l_0 + 2l_2: \{-y < 2 \cdot x, 2 \cdot x < -2\} \vdash_{\text{LRA}} -y < -2$
Deduce: $l_3: -y < -2$ (level 0) /* late propagation */
3. $y \leftarrow 0 \vdash_{\text{LRA}} \overline{-y < -2}$ detects LRA-conflict $\{y \leftarrow 0, -y < -2\}$
UndoClear: undo $?(y \leftarrow 0)$ and back to level 0
4. Decide: $?(x \leftarrow -2)$ (level 1) /* acceptable */
LRA-conflict: $\{x + y < 0, -y < -2, x \leftarrow -2\}$
5. Explained by $l_1 + l_3: \{x < -y, -y < -2\} \vdash_{\text{LRA}} x < -2$
Deduce: $l_4: x < -2$ (level 0) /* late propagation */

The non-terminating LRA example in CDSAT

6. $x \leftarrow -2 \vdash_{\text{LRA}} \overline{x < -2}$ detects **LRA-conflict** $\{x \leftarrow -2, x < -2\}$
UndoClear: undo $? (x \leftarrow -2)$ and back to level 0
7. **Decide**: $? (y \leftarrow -3)$ (level 1) /* **acceptable** */
LRA-conflict: $\{-2 \cdot x - y < 0, x < -2, y \leftarrow -3\}$
8. Explained by $l_0 + 2l_4$: $\{-y < 2 \cdot x, 2 \cdot x < -4\} \vdash_{\text{LRA}} -y < -4$
Deduce: $l_5: -y < -4$ (level 0) /* **late propagation** */
9. $y \leftarrow -3 \vdash_{\text{LRA}} \overline{-y < -4}$ detects **LRA-conflict** $\{y \leftarrow -3, -y < -4\}$
UndoClear: undo $? (y \leftarrow -3)$ and back to level 0
10. **Decide**: $? (x \leftarrow -3)$ (level 1) /* **acceptable** */
LRA-conflict: $\{x + y < 0, -y < -4, x \leftarrow -3\}$

The non-terminating LRA example in CDSAT

11. Explained by $l_1 + l_5: \{x < -y, -y < -4\} \vdash_{\text{LRA}} x < -4$
Deduce: $l_6: x < -4$ (level 0) /* late propagation */
12. $x \leftarrow -3 \vdash_{\text{LRA}} \overline{x < -4}$ detects LRA-conflict $\{x \leftarrow -3, x < -4\}$
UndoClear: undo $? (x \leftarrow -3)$ and back to level 0
13. Decide: $? (y \leftarrow -5)$ (level 1) /* acceptable */
LRA-conflict: $\{-2 \cdot x - y < 0, x < -4, y \leftarrow -5\}$
14. Explained by $l_0 + 2l_6: \{-y < 2 \cdot x, 2 \cdot x < -8\} \vdash_{\text{LRA}} -y < -8$
Deduce: $l_7: -y < -8$ (level 0) /* late propagation */
15. $y \leftarrow -5 \vdash_{\text{LRA}} \overline{-y < -8}$ detects LRA-conflict $\{y \leftarrow -5, -y < -8\}$
UndoClear: undo $? (y \leftarrow -5)$ and back to level 0
- ...

Solution of the non-termination problem

- ▶ Assume a **fixed total ordering** $\prec_{\text{LRA}}$ on rational variables
- ▶ Apply FM-resolution only if the variable resolved upon is $\prec_{\text{LRA}}$ -**maximum** in both premises
- ▶ Solution adopted in:
 - ▶ Conflict Resolution [Korovin, Tsiskaridze, Voronkov: CP 2009]
 - ▶ GDPLL aka GCDCL [McMillan, Kuehlmann, Sagiv: CAV 2009]
 - ▶ The **LRA**-module in CDSAT

Completing the CDSAT module for LRA

- ▶ **FM-resolution**: $\{t_1 \triangleleft_1 x, x \triangleleft_2 t_2\} \vdash_{\text{LRA}} t_1 \triangleleft_3 t_2$
 $\triangleleft_1, \triangleleft_2, \triangleleft_3 \in \{<, \leq\}$
 $\triangleleft_3$ is $<$ if either $\triangleleft_1$ or $\triangleleft_2$ is $<$ and $\leq$ otherwise
- ▶ **basis_{LRA}(X)**: subterms, equalities, disequalities generated by the inference rules
for FM-resolution only the disequalities that FM-resolution generates by resolving on the $\prec_{\text{LRA-maximum variable}}$ in the premises are included

Completing the CDSAT module for LRA (I version)

- Detection of an empty solution space:

$$\{y_1 \leftarrow \tilde{q}_1, \dots, y_m \leftarrow \tilde{q}_m\} \uplus E \vdash_{\text{LRA}} \perp$$

if for all x in E there is an i ($1 \leq i \leq m$) such that

$x \prec_{\text{LRA}} y_i$ or $x = y_i$ (1st horn for at least one x)

and $\{y_1 \leftarrow \tilde{q}_1, \dots, y_m \leftarrow \tilde{q}_m\} \uplus E$ is LRA-unsatisfiable

- No value for some x given the assignments to $y_1, \dots, y_m$

Completing the CDSAT module for LRA (II version)

- ▶ Omit **detection of an empty solution space**
- ▶ An LRA-assignment J is **sensible** if $x \prec_{\text{LRA}} y$ and J assigns a value to y imply that J assigns a value to x
- ▶ Adopt a **sensible** search plan: selects rational variables for decision in **$\prec_{\text{LRA}}$ -increasing order**
- ▶ LRA-assignments generated by a **sensible** search plan are **sensible**

The non-terminating example fixed in CDSAT

- ▶ Assume $y \prec_{\text{LRA}} x$
- ▶ 2nd FM-resolution inference in the non-halting sequence:
 $\{x < -y, -y < -2\} \vdash_{\text{LRA}} x < -2$
is barred: it resolves on y when x occurs in the premises
- ▶ All derivations embedding that diverging series of FM-resolution inferences are barred
- ▶ Multiple CDSAT-derivations discover that
 $l_0: -2 \cdot x - y < 0, l_1: x + y < 0, l_2: x < -1$
is **LRA-unsatisfiable**
- ▶ A simple one does it by mere **LRA-propagations** at level 0

The non-terminating example fixed in CDSAT

$l_0: -2 \cdot x - y < 0$, $l_1: x + y < 0$, $l_2: x < -1$ (level 0)

Assume $y \prec_{\text{LRA}} x$

1. **Deduce:** $l_3: -y < -2$ (level 0)

$l_0 + 2l_2: \{-y < 2 \cdot x, 2 \cdot x < -2\} \vdash_{\text{LRA}} -y < -2$

/* x is $\prec_{\text{LRA}}$ -max variable in both premises */

2. **Deduce:** $l_4: y < 0$ (level 0) /*normal form of $-y < -2 \cdot y$ */

$l_0 + 2l_1: \{-y < 2 \cdot x, 2 \cdot x < -2 \cdot y\} \vdash_{\text{LRA}} -y < -2 \cdot y$

/* x is $\prec_{\text{LRA}}$ -max variable in both premises */

The non-terminating example fixed in CDSAT

$l_0: -2 \cdot x - y < 0$, $l_1: x + y < 0$, $l_2: x < -1$ (level 0)

Assume $y \prec_{\text{LRA}} x$

3. **Deduce**: $l_5: 2 < 0$ (level 0)
 $-l_3 + l_4: \{2 < y, y < 0\} \vdash_{\text{LRA}} 2 < 0$
/* y is $\prec_{\text{LRA}}$ -max variable in both premises as there is no x^* */
4. $\emptyset \vdash_{\text{LRA}} \overline{2 < 0}$ reveals **conflict** at level 0:
Fail returns **unsat**

Outline of GCDCL procedure for generic single theory $\mathcal{T}$

- ▶ How about going from $\mathcal{T}$ -literals to $\mathcal{T}$ -clauses?
- ▶ Embed reasoning about disjunction into theory reasoning by generalizing to $\mathcal{T}$ -clauses a theory reasoning inference rule for $\mathcal{T}$ -literals
- ▶ Apply the generalized rule only to **explain conflicts**
- ▶ Devise restrictions to ensure **termination**

Achieved in GCDCL: only linear rational arithmetic (**LRA**)

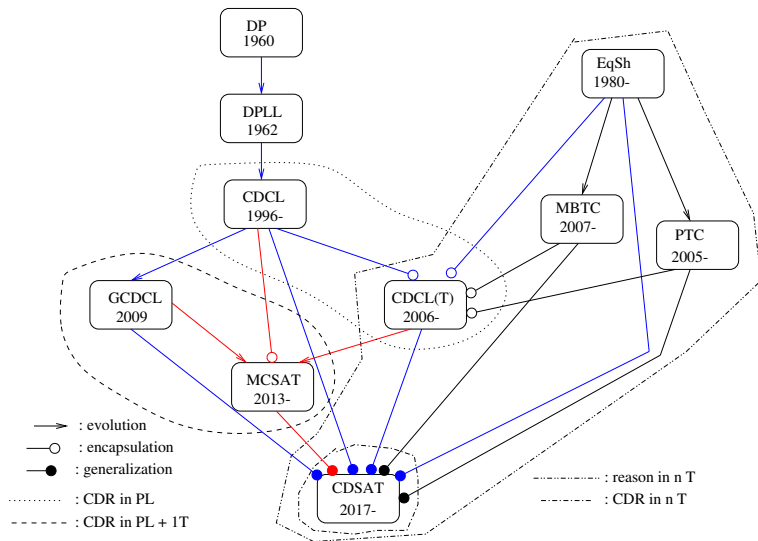
[McMillan, Kuehlmann, Sagiv: CAV 2009]

Generalized CDCL (GCDCL) for LRA

- ▶ Input: set S of LRA-clauses
- ▶ LRA-clause: disjunction of LRA-literals
- ▶ Generalize FM-resolution to LRA-clauses: shadow rule e.g.:
 $\{(b < d) \vee (c < d), d < a\} \vdash_{\text{LRA}} (b < a) \vee (c < a)$
- ▶ Generates new (non-input) atoms
- ▶ Applied only to explain LRA-conflicts
generating lemmas excluding LRA-assignments
- ▶ Add restrictions to ensure termination

[McMillan, Kuehlmann, Sagiv: CAV 2009]

The big picture: better conflict-driven one-theory reasoning



MCSAT (Model-Constructing SATisfiability)

- ▶ Entrust the reasoning about disjunction to CDCL
- ▶ Integrate in CDCL a conflict-driven $\mathcal{T}$ -satisfiability procedure for sets of $\mathcal{T}$ -literals
- ▶ CDCL($\mathcal{T}$)?
 - No, it allows neither **first-order assignment**
 - nor **new** atoms on the trail
 - nor **$\mathcal{T}$ -inferences** generating lemmas excluding $\mathcal{T}$ -assignments

[de Moura, Jovanović: VMCAI 2013], [Jovanović, Barrett, de Moura: FMCAD 2013], [Jovanović: VMCAI 2017], [Bobot, Graham-Lengrand, Marre, Bury: SMT 2018], [Graham-Lengrand, Jovanović, Dutertre: IJCAR 2020]

MCSAT (Model-Constructing SATisfiability)

- ▶ Integrate CDCL and **one model-constructing** conflict-driven $\mathcal{T}$ -sat procedure for sets of $\mathcal{T}$ -literals (**$\mathcal{T}$ -plugin**) that
 - ▶ Has access to the trail
 - ▶ Proposes assignments to first-order terms: $\mathcal{T}$ -assignment
 - ▶ Computes **$\mathcal{T}$ -propagations**
 - ▶ **Explains $\mathcal{T}$ -conflicts** by **$\mathcal{T}$ -inferences** generating lemmas excluding $\mathcal{T}$ -assignments
 - ▶ Lemma may contain **new** (i.e., non-input) atoms coming from a **finite basis** for **termination**
- ▶ CDCL and the **$\mathcal{T}$ -plugin** cooperate in model construction
- ▶ Both **propositional** and **$\mathcal{T}$ -reasoning** are **conflict-driven**

CDSAT generalizes MCSAT

- ▶ CDSAT generalizes MCSAT to a generic union $\mathcal{T} = \bigcup_{k=1}^n \mathcal{T}_k$
- ▶ MCSAT is **not** a combination calculus
hence does not cover, e.g.:
 - ▶ Interaction of multiple first-order theories on the trail
 - ▶ Conflict-drivenness for more than one first-order theory
 - ▶ Combination of conflict-driven and **black-box** procedures
 - ▶ **Soundness**, **completeness**, **termination** for theory combination
 - ▶ Construction of **finite global basis** from local ones
- ▶ CDSAT does **not** require model-constructing $\mathcal{T}_k$ -sat procedures in MCSAT arith-inspired strong sense

CDSAT generalizes MCSAT

- ▶ The CDSAT and MCSAT transition systems are different
- ▶ What MCSAT does by model-evaluation mechanism and theory-explanation function is done in CDSAT by $\mathcal{T}_k$ -inferences
 - ▶ MCSAT: explanation function **private** to $\mathcal{T}$ -plugin
 - ▶ CDSAT: $\mathcal{T}_k$ -inferences **public** on **shared** trail
- ▶ MCSAT: integration of CDCL and a $\mathcal{T}$ -plugin
CDSAT: reasoning in the **union** of Bool and $\mathcal{T}$

CDSAT emulates MCSAT if the theory union contains only Bool and **one** theory $\mathcal{T}$ equipped with a conflict-driven model-constructing $\mathcal{T}$ -sat procedure for sets of $\mathcal{T}$ -literals

Example where CDSAT emulates MCSAT

- ▶ $H = \{x < y, x < z, (y < w) \vee (z < w), w < x\}$
- ▶ Theory union: **LRA** and **Bool** sharing sort prop
- ▶ **Bool**: $<$ is a foreign symbol, the $<$ -terms are foreign terms seen as propositional variables
- ▶ **LRA**: $\vee$ is a foreign symbol, the disjunction is a foreign term seen as a propositional variable
- ▶ $\mathcal{V}_{\text{sh}}(H) = \{x < y, x < z, (y < w) \vee (z < w), w < x, y < w, z < w\}$

Example where CDSAT emulates MCSAT

$x < y, x < z, (y < w) \vee (z < w), w < x$ (level 0)

LRA-module:

- ▶ Ordering on rational variables: $x \prec_{\text{LRA}} y \prec_{\text{LRA}} z \prec_{\text{LRA}} w$
- ▶ Sensible search plan

1. Decide: $?(x \leftarrow 0)$ (level 1) /* acceptable for LRA */
2. Decide: $?(y \leftarrow 1)$ (level 2) /* acceptable for LRA */
/* $?(y \leftarrow 0)$ not acceptable for LRA: $\{x \leftarrow 0, y \leftarrow 0\} \vdash_{\text{LRA}} \overline{(x < y)}$ */
3. Decide: $?(z \leftarrow 1)$ (level 3) /* acceptable for LRA */
/* $?(z \leftarrow 0)$ not acceptable for LRA: $\{x \leftarrow 0, z \leftarrow 0\} \vdash_{\text{LRA}} \overline{(x < z)}$ */

Example where CDSAT emulates MCSAT

4. **Decide:** $\exists(y < w)$ (level 4) /* **acceptable** for **Bool***/
5. **Deduce:** $J \vdash (y < x)$ with $J = \{y < w, w < x\}$ (level 4)
 $\{y < w, w < x\} \vdash_{\text{LRA}} y < x$
/* w is $\prec_{\text{LRA-max}}$ variable in both $y < w$ and $w < x$ */
6. **Deduce:** $I \vdash (x < x)$ with $I = \{x < y, y < x\}$ (level 4)
 $\{x < y, y < x\} \vdash_{\text{LRA}} x < x$
/* y is $\prec_{\text{LRA-max}}$ variable in both $x < y$ and $y < x$ */
LRA-conflict: $E_0 = \{x < x\}$ (level 4)
7. **Resolve:** $E_1 = \{x < y, y < x\}$
8. **Resolve:** $E_2 = \{x < y, y < w, w < x\}$

Example where CDSAT emulates MCSAT

$E_2 = \{x < y, y < w, w < x\}$: **outstanding** in E_2
 $\text{level}_\Gamma(y < w) = 4, \text{level}_\Gamma(x < y) = \text{level}_\Gamma(w < x) = 0$

9. **Backjump**: back to level 0 adding $\kappa \vdash (\overline{y < w})$

$K = \{x < y, w < x\}$

10. **Deduce**: $G \vdash (z < w)$ (level 0)

$G = \{\overline{y < w}, (y < w) \vee (z < w)\}$ (level 0)

$\{\overline{y < w}, (y < w) \vee (z < w)\} \vdash_{\text{Bool}} z < w$

/* shadow rule unnecessary: **Bool**-module handles disjunction

LRA-module reasons about **LRA**-literals */

Example where CDSAT emulates MCSAT

11. **Deduce:** $M \vdash (z < x)$ with $M = \{z < w, w < x\}$ (level 0)
 $\{z < w, w < x\} \vdash_{\text{LRA}} z < x$
/* w is $\prec_{\text{LRA}}$ -max variable in both $z < w$ and $w < x$ */
12. **Deduce:** $N \vdash (x < x)$ with $N = \{x < z, z < x\}$ (level 0)
 $\{x < z, z < x\} \vdash_{\text{LRA}} x < x$
/* z is $\prec_{\text{LRA}}$ -max variable in both $x < z$ and $z < x$ */
13. **LRA-conflict:** $E_3 = \{x < x\}$ (level 0)
Fail returns **unsat**

CDSAT: Conflict-driven reasoning from a theory to many

- ▶ **Conflict-driven** behavior and **black-box** integration are at odds: each conflict-driven $\mathcal{T}_k$ -sat procedure needs to access the trail, post assignments, perform inferences, explain $\mathcal{T}_k$ -**conflicts**, post lemmas as assignments
- ▶ Key abstraction in CDSAT: open the **black-boxes** pull out the $\mathcal{T}_k$ -**inference systems** and coordinate them in a **conflict-driven** way
- ▶ If $\mathcal{T}_k$ has no conflict-driven $\mathcal{T}_k$ -sat procedure:
black-box inference rule $L_1, \dots, L_m \vdash_k \perp$
invokes the $\mathcal{T}_k$ -procedure to detect $\mathcal{T}_k$ -**unsat**

Reasoning in a theory union

$$\mathcal{T} = \bigcup_{i=1}^n \mathcal{T}_k$$

- ▶ Each component theory $\mathcal{T}_k$ has
 - ▶ Signature $\Sigma_k = (S_k, F_k)$ where S_k is the set of sorts and F_k is the set of symbols
 - ▶ **Conservative** extension $\mathcal{T}_k^+$ providing $\mathcal{T}_k$ -values
- ▶ The union theory $\mathcal{T}$ has all sorts, all symbols, all values:
 - ▶ Signature $\Sigma = (S, F)$ where $S = \bigcup_{i=1}^n S_k$ and $F = \bigcup_{i=1}^n F_k$
 - ▶ **Conservative** extension $\mathcal{T}^+ = \bigcup_{i=1}^n \mathcal{T}_k^+$

Key notions for conflict-driven reasoning in a theory union

$$\mathcal{T} = \bigcup_{i=1}^n \mathcal{T}_k$$

- ▶ Theory **view**: the view that a component theory $\mathcal{T}_k$ has of a $\mathcal{T}$ -assignment
- ▶ Assignments and models: **endorsement**
- ▶ **Relevance** of a term to a theory
- ▶ **Acceptability** of an assignment for a theory module in the presence of more than one theory

Theory view of an assignment

$$\mathcal{T} = \bigcup_{k=1}^n \mathcal{T}_k$$

- ▶ The $\mathcal{T}_k$ -view of a $\mathcal{T}$ -assignment H , denoted H_k , is the $\mathcal{T}_k$ -assignment that contains:
 - ▶ $\mathcal{T}_k$ -assignments in H : those that assign $\mathcal{T}_k$ -values
 - ▶ $u \simeq t$ if H contains $u \leftarrow c$ and $t \leftarrow c$ of a $\mathcal{T}_k$ sort s ($s \neq \text{prop}$)
 - ▶ $u \not\simeq t$ if H contains $u \leftarrow c$ and $t \leftarrow q$ of a $\mathcal{T}_k$ sort s ($s \neq \text{prop}$) with $c \neq q$
- ▶ The assignments $u \leftarrow c$ and $t \leftarrow c$ (or $t \leftarrow q$) may have been posted by another theory $\mathcal{T}_j$ ($k \neq j$) that also has sort s
- ▶ A Boolean assignment belongs to every theory view

Theory view in the CDSAT as MCSAT example

- ▶ Input: $x < y$, $x < z$, $(y < w) \vee (z < w)$, $w < x$
- ▶ Theories: **LRA** and **Bool**
- ▶ All input assignments are **Boolean**: in both theory views
- ▶ **First-order decisions** by **LRA** such as $?(x \leftarrow 0)$ and $?(y \leftarrow 1)$ and their consequences such as $x \neq y$: only in the **LRA**-view
- ▶ **Boolean decisions**: in both theory views

The view of the union theory: global view

$$\mathcal{T} = \bigcup_{k=1}^n \mathcal{T}_k$$

- ▶ The $\mathcal{T}$ -view of a $\mathcal{T}$ -assignment H , denoted $H_{\mathcal{T}}$, has everything:
 - ▶ Since $\mathcal{T}$ has all the values, all assignments are $\mathcal{T}$ -assignments and hence are in $H_{\mathcal{T}}$
 - ▶ Since $\mathcal{T}$ has all the sorts, all equalities and disequalities gleaned from first-order assignments are in $H_{\mathcal{T}}$
- ▶ The $\mathcal{T}$ -view is called **global view**

In order to see another example let's introduce another theory

A theory of a data structure: arrays

- ▶ Signature $\Sigma_{\text{Arr}} = (S, F)$ where:
 - ▶ Sorts: $S = \{\text{prop}, I, V, A\}$
 - ▶ I : sort of **indices**
 - ▶ V : sort of (array) **values**
 - ▶ A : sort of **arrays** with indices of sort I and values of sort V
 - ▶ Symbols: $F = \{\simeq_s \mid s \in S\} \cup \{\text{select}, \text{store}\}$
 - ▶ $\text{select}: A \times I \rightarrow V$ (read)
 - ▶ $\text{store}: A \times I \times V \rightarrow A$ (write)
- ▶ Theory **extension** Arr^+ : **trivial** or **countable**
(same choice as for **EU**F)

Theory view: 3-theory example

$$H = \{x > 1, \text{store}(a, i, v) \simeq b, \text{select}(a, j) \leftarrow \text{red}, y \leftarrow -1, z \leftarrow 2\}$$

- ▶ Theories: **Bool**, **Arr**, and **EUF** where
EUf has the sort Q of the rational numbers
red is an **Arr**-value of sort V
- ▶ $H_{\text{Bool}} = \{x > 1, \text{store}(a, i, v) \simeq b\}$
- ▶ $H_{\text{Arr}} = \{x > 1, \text{store}(a, i, v) \simeq b, \text{select}(a, j) \leftarrow \text{red}\}$
- ▶ $H_{\text{LRA}} = \{x > 1, \text{store}(a, i, v) \simeq b, y \leftarrow -1, z \leftarrow 2, y \neq z\}$
- ▶ $H_{\text{EUf}} = \{x > 1, \text{store}(a, i, v) \simeq b, y \neq z\}$
- ▶ **Global view:** $H \cup \{y \neq z\}$

Assignments and models: endorsement

- ▶ Let $\mathcal{M}$ be a $\mathcal{T}^+$ -model
- ▶ $\mathcal{M}$ **endorses** $u \leftarrow c$, written $\mathcal{M} \models (u \leftarrow c)$:
 $\mathcal{M}$ interprets term u and $\mathcal{T}$ -value c as the same element in $\mathcal{M}$'s domain for their sort
- ▶ **Boolean assignment**:
 - ▶ Endorsement is the same as formula satisfaction:
 $\mathcal{M} \models (I \leftarrow \text{true})$ iff $\mathcal{M} \models I$ and $\mathcal{M} \models (I \leftarrow \text{false})$ iff $\mathcal{M} \models \bar{I}$
 - ▶ No difference btw $\mathcal{T}$ -model and $\mathcal{T}^+$ -model
- ▶ **First-order assignment**: $\mathcal{T}^+$ -model needed to interpret $\mathcal{T}$ -values and interpret them consistently with $\mathcal{T}^+$ -axioms

Endorsement and theory view in one theory

- ▶ Let $\mathcal{M}$ be a $\mathcal{T}_k^+$ -model
- ▶ $\mathcal{M} \models \{u \leftarrow c, t \leftarrow c\}$ implies $\mathcal{M} \models (u \simeq t)$
- ▶ Let $J = \{u \leftarrow c, t \leftarrow q\}$ where $c \neq q$
 - ▶ $\mathcal{M} \models J$ does not imply $\mathcal{M} \models (u \not\simeq t)$
 - ▶ $\mathcal{M} \models J_{\mathcal{T}_k}$ implies $\mathcal{M} \models (u \not\simeq t)$
- ▶ The desired property is **endorsement of the theory view**

Satisfiability of an assignment in one theory

- ▶ A $\mathcal{T}_k$ -assignment J is **satisfiable** if there exists a $\mathcal{T}_k^+$ -model $\mathcal{M}$ such that $\mathcal{M} \models J_{\mathcal{T}_k}$
- ▶ $J \models L$: for all $\mathcal{T}_k^+$ -models $\mathcal{M}$
if $\mathcal{M} \models J_{\mathcal{T}_k}$ then $\mathcal{M} \models L$
- ▶ **Sound** $\mathcal{T}_k$ -inference: if $J \vdash_k L$ then $J \models L$

$$\mathcal{T} = \bigcup_{k=1}^n \mathcal{T}_k$$

- ▶ A $\mathcal{T}$ -assignment H is **satisfiable** if there exists a $\mathcal{T}^+$ -model $\mathcal{M}$ such that $\mathcal{M} \models H_{\mathcal{T}}$
- ▶ A $\mathcal{T}^+$ -model may identify values from different theories:
 - ▶ $H = \{t \leftarrow 3.1, u \leftarrow 5.4, t \leftarrow \text{red}, u \leftarrow \text{blue}\}$
 - ▶ $\mathcal{T}_i^+$ -model $\mathcal{M}_i$: $\mathcal{M}_i \models \{t \leftarrow 3.1, u \leftarrow 5.4, t \neq u\}$
 - ▶ $\mathcal{T}_j^+$ -model $\mathcal{M}_j$: $\mathcal{M}_j \models \{t \leftarrow \text{red}, u \leftarrow \text{blue}, t \neq u\}$
 - ▶ $\mathcal{T}^+$ -model $\mathcal{M}$: $\mathcal{M} \models H \cup \{t \neq u\}$
interpreting 3.1 and red as the same $\mathcal{M}$ -element e_1
interpreting 5.4 and blue as the same $\mathcal{M}$ -element e_2
with $e_1 \neq e_2$

Towards relevance: terms occurring in an assignment

- ▶ $\sqsubseteq$ is the subterm relation
- ▶ Given assignment $J = \{u_1 \leftarrow c_1, \dots, u_m \leftarrow c_m\}$
- ▶ The set $G(J)$ of the terms that **occur** in J contains the assigned terms and their subterms:
$$G(J) = \{t \mid t \sqsubseteq u_i, 1 \leq i \leq m\}$$

Relevance for disjoint theories

Term u is **relevant** to $\mathcal{T}_k$ in a $\mathcal{T}_k$ -assignment J if

- ▶ Either u occurs in J (i.e., $u \in G(J)$) and $\mathcal{T}_k$ has the sort s of u and has $\mathcal{T}_k$ -values for s
- ▶ Or term u is an equality $u_1 \simeq_s u_2$ such that
 - ▶ u_1 and u_2 occur in assignment J (i.e., $u_1 \in G(J)$, $u_2 \in G(J)$)
 - ▶ $\mathcal{T}_k$ has sort s
 - ▶ $\mathcal{T}_k$ does **not** have $\mathcal{T}_k$ -values of sort s

Either way $\mathcal{T}_k$ can propose an assignment for term u

Relevance and equality

Relevance determines how different theories make decisions about equalities:

- ▶ Suppose that theory $\mathcal{T}$ has sort s
- ▶ If $\mathcal{T}$ has $\mathcal{T}$ -values for s (i.e., $\mathcal{T}^+$ adds constants of sort s), it can decide whether two terms u_1 and u_2 of sort s are equal by assigning them equal or different values:
 $\{u_1 \leftarrow c, u_2 \leftarrow c\}$ or $\{u_1 \leftarrow c, u_2 \leftarrow q\}$
- ▶ Otherwise, $\mathcal{T}$ can decide whether u_1 and u_2 are equal by assigning a **Boolean value** to $u_1 \simeq u_2$:
 $(u_1 \simeq u_2) \leftarrow \text{true}$ or $(u_1 \simeq u_2) \leftarrow \text{false}$

Relevance: example

- ▶ $H = \{x \leftarrow 5, f(x) \leftarrow 2, f(y) \leftarrow 3\}$
- ▶ $x, y: Q, \quad f: Q \rightarrow Q, \quad \text{LRA and EUF share sort } Q$
- ▶ $H_{\text{LRA}} = H \cup \{x \neq f(x), x \neq f(y), f(x) \neq f(y)\}$
- ▶ $H_{\text{EUF}} = \{x \neq f(x), x \neq f(y), f(x) \neq f(y)\}$
- ▶ x and y are **LRA-relevant**, not **EUF-relevant**
- ▶ $x \simeq y$ is **EUF-relevant**, not **LRA-relevant**
- ▶ **LRA** makes x and y equal/different by assigning them same/different values
- ▶ **EUF** makes x and y equal/different by assigning a truth value to $x \simeq y$

Relevance for predicate-sharing theories

Term u is **relevant** to $\mathcal{T}_k$ in a $\mathcal{T}_k$ -assignment J if

- ▶ Either $u \in G(J)$ and $\mathcal{T}_k$ has the sort s of u and $\mathcal{T}_k$ has $\mathcal{T}_k$ -values for s
- ▶ Or term u is $u_1 \simeq_s u_2$ such that $u_1 \in G(J)$, $u_2 \in G(J)$, $\mathcal{T}_k$ has sort s , but does **not** have $\mathcal{T}_k$ -values of sort s
- ▶ Or u is a Boolean term $p(u_1, \dots, u_m)$ such that $p: (s_1 \times \dots \times s_m) \rightarrow \text{prop}$ is a **shared predicate symbol** by $\mathcal{T}_k$ and at least another theory $\mathcal{T}_j$ and for all i ($1 \leq i \leq m$) $u_i \in G(J)$ and $\mathcal{T}_k$ has sort s_i

The CDSAT transition system assumes a global basis

- ▶ Basic transition system:
 - ▶ Trail rules: Decide, Deduce, Fail, ConflictSolve
 - ▶ Conflict state rules: Resolve, Backjump, UndoClear, UndoDecide
- ▶ Standard transition system:
 - ▶ Trail rules: Decide, Deduce, Fail, ConflictSolve
 - ▶ Conflict state rules: Resolve, LearnBackjump, UndoClear, UndoDecide
- ▶ Parameter: global basis $\mathcal{B}$:
 - ▶ A set from which CDSAT can draw new terms
 - ▶ Used to prove termination of CDSAT
 - ▶ It can be constructed from the local bases

Trail rule Decide and acceptability revisited

$\Gamma_{\mathcal{T}_k}$: the $\mathcal{T}_k$ -view of trail Γ

Decide: $\Gamma \longrightarrow \Gamma, ?(u \leftarrow c)$

if $u \leftarrow c$ is an **acceptable** $\mathcal{T}_k$ -assignment for the $\mathcal{T}_k$ -module $\mathcal{I}_k$ in Γ_k :

- ▶ Term u is **relevant** to $\mathcal{T}_k$ in $\Gamma_{\mathcal{T}_k}$
- ▶ $\Gamma_{\mathcal{T}_k}$ does not assign a $\mathcal{T}_k$ -value to term u
- ▶ If $u \leftarrow c$ is a first-order assignment: for no $J \subseteq \Gamma_{\mathcal{T}_k}$ there exists an $\mathcal{I}_k$ -inference $J \cup \{u \leftarrow c\} \vdash_{\mathcal{T}_k} L$ and $\bar{L} \in \Gamma_{\mathcal{T}_k}$

Trail rule Deduce revisited

- ▶ Transition rule **Deduce** adds to the trail a Boolean **justified assignment** $J \vdash L$:

$$\text{Deduce: } \Gamma \longrightarrow \Gamma, J \vdash L$$

if a $\mathcal{T}_k$ -module $\mathcal{I}_k$ has inference $J \vdash_k L$ for $J \subseteq \Gamma$
and $L \notin \Gamma$ and $\bar{L} \notin \Gamma$

- ▶ L is in the **global basis** $\mathcal{B}$
- ▶ **Deduce** covers both $\mathcal{T}_k$ -propagation and preliminary explanation of $\mathcal{T}_k$ -**conflicts**

Explanation of conflicts in CDSAT: recap

- ▶ Preliminary explanation of a $\mathcal{T}_k$ -conflict by $\mathcal{T}_k$ -inferences encapsulated as Deduce steps: not in conflict state
- ▶ Until the conflict surfaces as a Boolean conflict:
 $J \vdash_k L$ and $\bar{L} \in \Gamma$: $H = J \cup \{\bar{L}\}$ is a conflict
- ▶ Switch to conflict state $\langle \Gamma; H \rangle$
- ▶ Explanation of conflict H by replacing justified assignments in H with their justifications: Resolve transition rule

Conflict state rule Resolve revisited

- ▶ Conflict state: $\langle \Gamma; E \uplus \{A\} \rangle$ and $H \vdash A \in \Gamma$
- ▶ Transition rule **Resolve explains** the conflict by replacing $H \vdash A$ with its justification H :

$$\text{Resolve: } \langle \Gamma; E \uplus \{A\} \rangle \Longrightarrow \langle \Gamma; E \cup H \rangle$$

- ▶ Provided H does not contain a **first-order decision** A' such that $\text{level}_\Gamma(A') = \text{level}_\Gamma(E \uplus \{A\})$ /* case covered by **UndoDecide** */
- ▶ A can be a **Boolean** or a **first-order** assignment: if first-order, then input ($H = \emptyset$) and removed from conflict, not from Γ

Conflict state rule Backjump revisited

- Conflict state: $\langle \Gamma; E \uplus \{L\} \rangle$ where assignment L is **outstanding** in $E \uplus \{L\}$, and $\text{level}_\Gamma(E) = m$
- Transition rule **Backjump** solves the conflict by jumping back to level m and adding $E \vdash \bar{L}$ to the trail:

$$\text{Backjump: } \langle \Gamma; E \uplus \{L\} \rangle \Longrightarrow \Gamma^{\leq m}, E \vdash \bar{L}$$

- **Bool** only: $\text{level}_\Gamma(L)$ is the current level, the trail is a stack
Theory union: L may be a **late propagation**, $\text{level}_\Gamma(L)$ may be smaller than the current one, the trail is not a stack

Conflict state rule LearnBackjump revisited

- Conflict state: $\langle \Gamma; E \uplus H \rangle$ where H contains only **Boolean assignments**: $H = \{\bar{L}_1, \dots, \bar{L}_k\}$
- Let $L = L_1 \vee \dots \vee L_k$: **clausal form** of H
- Transition rule **LearnBackjump** solves the conflict by jumping back to a level m such that $\text{level}_\Gamma(E) \leq m < \text{level}_\Gamma(H)$ and adding ${}_E \vdash L$:

$$\text{LearnBackjump: } \langle \Gamma; E \uplus H \rangle \Longrightarrow \Gamma^{\leq m}, {}_E \vdash L$$

- Provided $L \notin \Gamma$, $\bar{L} \notin \Gamma$, and L is in the **global basis** $\mathcal{B}$

Conflict state rule UndoDecide

- ▶ Conflict state: $\langle \Gamma; E \uplus \{L\} \rangle$ where
 - ▶ L is a **justified assignment** $H \vdash L$
 - ▶ Justification H contains a **first-order decision** A
 - ▶ $\text{level}_\Gamma(A) = \text{level}_\Gamma(H) = \text{level}_\Gamma(L) = \text{level}_\Gamma(E) = m$
 A determines the level of H (hence of L) and also that of E
- ▶ Transition rule **UndoDecide** solves the conflict by undoing A , removing all assignments of level greater than or equal to m , and posting $\bar{L}$ on the trail as a **decision**:

$$\text{UndoDecide: } \langle \Gamma; E \uplus \{L\} \rangle \Longrightarrow \Gamma^{\leq m-1}, ?\bar{L}$$

- ▶ A cannot be flipped but its **Boolean consequence** L can

Conflict state rule UndoDecide

- ▶ Decision $\bar{L}$ prevents repeating **first-order decision** A that caused the conflict, as $\bar{L}$ makes A **unacceptable**
- ▶ Similar to **UndoClear**, **UndoDecide** undoes a **first-order decision** and clears the trail of all assignments of equal or greater level
- ▶ Unlike **UndoClear**, and similar to **Backjump** or **LearnBackjump**, **UndoDecide** flips a **Boolean assignment**
- ▶ Unlike **Backjump** and **LearnBackjump**, the flipped **Boolean assignment** is a **decision**, not a **justified assignment**

Why is UndoDecide needed?

- ▶ UndoDecide applies when no other conflict state rule does
- ▶ UndoDecide applies to a conflict **without outstanding assignment**: neither Backjump nor UndoClear apply
- ▶ LearnBackjump generalizes Backjump by flipping a cube rather than a singleton, but it does not handle a conflict caused by a **first-order decision**
- ▶ Resolve must be forbidden:
replacing $H \vdash L$ with H in E and undoing $?A$ by UndoClear cause a loop if Decide reiterates $?A$
- ▶ Towards an example, let's continue with arrays

CDSAT module for the theory of arrays

The axiomatization of **Arr** features the **select-over-store** axioms:

1. $\forall a, v, i. \text{select}(\text{store}(a, i, v), i) \simeq v$
2. $\forall a, v, i, j. i \neq j \longrightarrow \text{select}(\text{store}(a, i, v), j) \simeq \text{select}(a, j)$

That yield inference rules in the module for **Arr**:

1. $\{i \simeq j, b \simeq \text{store}(a, i, v), \text{select}(b, j) \neq v\} \vdash_{\text{Arr}} \perp$
2. $\{i \neq j, b \simeq \text{store}(a, i, v), \text{select}(b, j) \neq \text{select}(a, j)\} \vdash_{\text{Arr}} \perp$

CDSAT module for the theory of arrays

- ▶ When going from axioms to rules, premises are
 - ▶ Flattened: e.g., replace $\text{select}(\text{store}(a, i, v), j)$ with $\text{select}(b, j)$ adding $b \simeq \text{store}(a, i, v)$, and
 - ▶ Linearized (no repeated variables): e.g., replace $\text{select}(\text{store}(a, i, v), i)$ with $\text{select}(\text{store}(a, i, v), j)$ adding $i \simeq j$
- ▶ So that they apply more generally:
 - ▶ Flattening and linearization do not require inferences
 - ▶ Replacement of equals by equals does
 - ▶ Equality rules do not have replacement:
only the **EUF** module reasons about congruence

CDSAT module for arrays with extensionality

- **Extensionality** axiom:

$$\forall a, b. (\forall i. \text{select}(a, i) \simeq \text{select}(b, i)) \longrightarrow a \simeq b$$

- Clausal form:

$$\text{select}(a, \text{diff}(a, b)) \not\simeq \text{select}(b, \text{diff}(a, b)) \vee a \simeq b$$

Skolem function $\text{diff}: A \times A \rightarrow I$ captures the witness index

- Inference rule:

$$a \not\simeq b \vdash_{\text{Arr}} \text{select}(a, \text{diff}(a, b)) \not\simeq \text{select}(b, \text{diff}(a, b))$$

- **basis_{Arr}(X)**: all subterms of terms in **X**, equalities btw them, and witness terms $\text{select}(a, \text{diff}(a, b))$, $\text{select}(b, \text{diff}(a, b))$

Example with theory clauses and UndoDecide

- ▶ Input set S contains clauses:
 - ▶ $C_1: (i \neq j) \vee (\text{select}(\text{store}(a, i, v), j) < \text{select}(a, j))$
 - ▶ $C_2: (\text{select}(a, j) - \text{select}(a, k)) \simeq 0$
 - ▶ $C_3: (\text{select}(\text{store}(a, i, v), j) \not< \text{select}(a, j)) \vee (\text{select}(a, j) + \text{select}(a, k) \simeq v)$
- ▶ Theory union: $\text{Bool} \cup \text{LRA} \cup \text{Arr}$
- ▶ Suppose Arr interprets indices as integers:
 $I = \mathbb{Z}$ and Arr^+ adds integer constants as Arr -values

Example with theory clauses and UndoDecide

1. Arr-module: Decide $\exists(i \leftarrow 0)$ (level 1)
/* acceptable as i is relevant to Arr */
2. Arr-module: Decide $\exists(j \leftarrow 0)$ (level 2)
3. Arr-module: equality inference $\{i \leftarrow 0, j \leftarrow 0\} \vdash_{\text{Arr}} i \simeq j$
Deduce: $A_1: \vdash (i \simeq j)$ with $J = \{i \leftarrow 0, j \leftarrow 0\}$ (level 2)
4. Bool-module: unit propagation
 $\{A_1, C_1\} \vdash_{\text{Bool}} \text{select}(\text{store}(a, i, v), j) < \text{select}(a, j)$
Deduce: $A_2: \vdash (\text{select}(\text{store}(a, i, v), j) < \text{select}(a, j))$
with $I = \{A_1, C_1\}$ (level 2)

Example with theory clauses and UndoDecide

5. Bool-module: unit propagation

$\{A_2, C_3\} \vdash_{\text{Bool}} \text{select}(a, j) + \text{select}(a, k) \simeq v$

Deduce: $A_3: H \vdash (\text{select}(a, j) + \text{select}(a, k) \simeq v)$

with $H = \{A_2, C_3\}$ (level 2)

6. Arr-module: first select-over-store rule

$\{A_1, A_2\} \vdash_{\text{Arr}} v < \text{select}(a, j)$

Deduce: $A_4: G \vdash (v < \text{select}(a, j))$ with $G = \{A_1, A_2\}$ (level 2)

7. LRA-module: FM-resolution $A_3 + C_2$

$\{A_3, C_2\} \vdash_{\text{LRA}} \text{select}(a, j) \simeq v/2$

Deduce: $A_5: M \vdash (\text{select}(a, j) \simeq v/2)$ with $M = \{A_3, C_2\}$ (level 2)

Example with theory clauses and UndoDecide

LRA-conflict: $E_0 = \{A_4, A_5\}$

as $A_4: \mathcal{G} \vdash (v < \text{select}(a, j))$ and $A_5: \mathcal{M} \vdash (\text{select}(a, j) \simeq v/2)$

8. E_0 contains literals A_4 and A_5 of max level (2)

Resolve: $E_1 = \{A_4, A_3, C_2\}$

9. E_1 contains literals A_3 and A_4 of max level (2)

Resolve: $E_2 = \{A_1, A_2, A_3, C_2\}$

10. E_2 contains literals A_1, A_2 and A_3 of max level (2)

Resolve: $E_3 = \{A_1, A_2, C_3, C_2\}$

11. E_3 contains literals A_1 , and A_2 of max level (2)

Resolve: $E_4 = \{A_1, C_1, C_3, C_2\}$

Example with theory clauses and UndoDecide

$$E_4 = \{A_1, C_1, C_3, C_2\}$$

E_4 contains an **outstanding** literal: $\text{level}_\Gamma(A_1) = 2 = \text{level}_\Gamma(E_4)$

Let $m = 2$

A_1 is $J \vdash (i \simeq j)$ and $J = \{i \leftarrow 0, j \leftarrow 0\}$

where $?(j \leftarrow 0)$ also has level $m = 2$

Apply **Resolve** to replace A_1 with J
and **UndoClear** to undo $?(j \leftarrow 0)$?

No, the system could loop by repeating $?(j \leftarrow 0)$
(still **acceptable** for **Arr**)

Example with theory clauses and UndoDecide

- 12. **UndoDecide**: undo $\gamma(j \leftarrow 0)$, go back to level $m - 1 = 1$,
and add **decision** $\gamma(i \neq j)$ (level 2)
/* clause C_1 is satisfied */
- 13. **LRA**-module: **Decide** $\gamma(\text{select}(a, j) \leftarrow 1)$ (level 3)
- 14. **LRA**-module: **Decide** $\gamma(\text{select}(a, k) \leftarrow 1)$ (level 4)
/* clause C_2 is satisfied */
- 15. **LRA**-module: **Decide** $\gamma(v \leftarrow 2)$ (level 5)
/* clause C_3 is satisfied */

Example with theory clauses and UndoDecide: variant

Suppose theory **Arr** does not have values for array indices:
 i and j not relevant to **Arr**, **Arr**-module cannot decide their values

1. **Arr**-module: **Decide** $?(i \simeq j)$ (level 1)
/* **acceptable** as $i \simeq j$ is relevant to **Arr** */
2. The same transitions as before lead to **conflict**
 $\{i \simeq j, C_1, C_3, C_2\}$ (level 1)
3. **Backjump** or **LearnBackjump** backtracks to level 0 and places
 $N \vdash (i \not\simeq j)$ on the trail with $N = \{C_1, C_3, C_2\}$
4. The satisfiability of the clauses can be detected as before

Three main theorems

- ▶ Input assignment: H_0
- ▶ All terms occurring in H are in **global basis** $\mathcal{B}$
- ▶ **CDSAT** is
 - ▶ **Sound**: if all theory modules are **sound**,
if CDSAT returns **unsat**, H_0 is unsatisfiable
 - ▶ **Terminating**: if $\mathcal{B}$ is **finite**,
CDSAT is guaranteed to terminate
 - ▶ **Complete**: if the leading theory module is **complete**
and the others are **leading-theory-complete**,
if CDSAT terminates without returning **unsat**,
there exists a $\mathcal{T}^+$ -model of Γ and hence of H_0

CDSAT for data structures

- ▶ Theories of data structures:
 - ▶ CDSAT allows to increase expressivity
 - ▶ Motivation for the extension of CDSAT to **predicate-sharing** theories
- ▶ **Arrays**, **maps**, and **vectors** with **abstract domain**

Arrays: finite of infinite?

In programming languages:

- ▶ Integer-indexed arrays
- ▶ **Finite**: indices in the interval $[0, n - 1]$, length n
- ▶ A **store** within bounds works, error/exception otherwise
- ▶ The computer memory is finite

Arrays: finite of infinite?

In the theory of arrays:

- ▶ All arrays have the same length:
the cardinality of the sort of indices
- ▶ If integer-indexed: **infinite** arrays
- ▶ No distinction btw in-bounds and out-of-bounds **store**

How about adding a length function len?

- ▶ Maps every array to its length: $\text{len}(a) \simeq n$
- ▶ Revised axiom of **extensionality** for integer-index arrays:
$$\forall a, b. [\text{len}(a) \simeq \text{len}(b) \wedge (\forall i. 0 \leq i < \text{len}(a) \rightarrow \text{select}(a, i) \simeq \text{select}(b, i))] \rightarrow a \simeq b$$
- ▶ Arrays and integers **no longer disjoint**
- ▶ Length is a **bridging function** and extensionality becomes a **bridging axiom**
- ▶ Most methods for reasoning in theory unions require disjoint theories (equality is the only shared symbol)

How about allowing quantifiers?

- ▶ Array property fragment
- ▶ Allows limited usage of $\forall$ over index variables, preserving decidability
- ▶ **Bounded array equality**: $beq(a, b, l, u)$ iff
 $\forall i. l \leq i \leq u \rightarrow \text{select}(a, i) \simeq \text{select}(b, i)$
- ▶ The theory of arrays and its limitations remain unchanged
- ▶ Efficient quantifier reasoning may be challenging

[Bradley, Manna, Sipma: VMCAI 2006]

Recurring to other theories? Sequences

- ▶ Using finite sequences with integer indices $[0, |x|)$ to model finite arrays
- ▶ **access/update** for **select/store**
- ▶ **Extensionality** axiom as in arrays with length
- ▶ **Update** axiom: update does not change length
only an update in $[0, |x|)$ modifies the element
- ▶ Conservative extension of the theory of integers
- ▶ Decidability of quantifier-free fragment: unknown
(sound and complete inference system, termination not guaranteed)

[Sheng et al.: IJCAR 2022, JAR 2023], [Ait-El-Hara, Bobot, Bury: 2025]

How about quantifiers and sequences together?

- ▶ The array property fragment with concatenation
- ▶ Arrays interpreted as finite integer-indexed sequences
- ▶ More expressive than the array property fragment: allows **index shifting** (e.g., $a[i]$ and $a[i + n]$)
concatenation can be defined
- ▶ Undecidable: halting problem of a two-register machine
- ▶ Decision procedure for **tangle-free** formulas
same as **stratified arrays**

[Wang, Appel: JAR 2023], [Ge, de Moura: CAV 2009]

Solution: a theory of arrays with abstract domain

- ▶ Neither quantifiers nor sequences:
enrich the theory of arrays with **length** and **admissibility**
- ▶ **Abstract domain** of definition of an array
indices not necessarily integers, nor linearly ordered
- ▶ **Admissibility** defined by another theory:
flexibility + minimum sharing
- ▶ Also **maps**, and **vectors** meaning **dynamic** arrays
- ▶ Theory unions including these theories are decidable:
from fitting the three theories in CDSAT

The theory of arrays with abstract domain: signature

- ▶ **ArrAD**: theory of arrays with abstract domain
- ▶ Sorts: indices I , values V , arrays A , lengths L , and prop
- ▶ **select**: $A \times I \rightarrow V$ **store**: $A \times I \times V \rightarrow A$ **len**: $A \rightarrow L$
- ▶ Free **admissibility** predicate: **Adm**: $I \times L \rightarrow \text{prop}$
Adm(i, l): index i is **admissible** wrt length l
- ▶ **Abstract domain**: definition of **Adm**
- ▶ **Concrete domain**: set of admissible indices given **Adm**'s definition and the interpretation of I
- ▶ **Adm** is **shared** with another theory $\mathcal{T}$ that defines it

The theory of arrays with abstract domain: axioms

► Select-over-store axioms:

- $\forall a, v, i. \text{select}(\text{store}(a, i, v), i) \simeq v$ is replaced by
 $\forall a, v, i. \text{Adm}(i, \text{len}(a)) \rightarrow \text{select}(\text{store}(a, i, v), i) \simeq v$
a store at an inadmissible index has no effect
- $\forall a, v, i, j. i \neq j \rightarrow \text{select}(\text{store}(a, i, v), j) \simeq \text{select}(a, j)$

► Store does **not** change length:

$$\forall a, i, v. \text{len}(\text{store}(a, i, v)) \simeq \text{len}(a)$$

► Extensionality:

$$\begin{aligned} &\forall a, b. [\text{len}(a) \simeq \text{len}(b) \wedge \\ &(\forall i. \text{Adm}(i, \text{len}(a)) \rightarrow \text{select}(a, i) \simeq \text{select}(b, i))] \\ &\rightarrow a \simeq b \end{aligned}$$

Example: the most common interpretation

- ▶ Let LIA be the theory defining **Adm**
- ▶ Indices and lengths are integers
- ▶ $\forall i, n. \text{Adm}(i, n) \leftrightarrow 0 \leq i < n$
- ▶ The **set of admissible indices** is the interval $[0, n)$
- ▶ Under this interpretation **extensionality** in ArrAD covers extensionality as in
 - ▶ Integer-indexed arrays with length
 - ▶ The theory of sequences
 - ▶ The array property fragment with concatenation

Example: capturing bounded equality

- ▶ Let LIA be the theory defining **Adm**
- ▶ Indices are integers, lengths are pairs of integers
- ▶ $\forall i, l, u. \text{Adm}(i, (l, u)) \leftrightarrow l \leq i \leq u$
- ▶ The **set of admissible indices** is the interval $[l, u]$
- ▶ Under this interpretation **extensionality** in ArrAD covers bounded equality as in the array property fragment

Example: length with starting address

- ▶ Let $\mathcal{T}$ be the theory defining **Adm**
- ▶ Indices are integers, length is a pair $(addr, n)$:
 - ▶ $addr$ (binary number): starting address of the array in memory
 - ▶ n (integer): the number of admissible indices
- ▶ $\forall i, addr, n. \text{Adm}(i, (addr, n)) \leftrightarrow 0 \leq i < n$
- ▶ The starting address does not affect admissibility but it affects array equality
- ▶ **Extensionality**: arrays a and b with same set of admissible indices, same values at all admissible indices, but different starting addresses are different (as in programming languages)

Example: admissibility as membership

- ▶ Let $\mathcal{T}$ be the theory defining **Adm**
- ▶ Indices are elements of a set S , length is a subsets of S
- ▶ $\forall i, N. \text{Adm}(i, N) \leftrightarrow i \in N$
- ▶ The **set of admissible indices** is the subset $N \subseteq S$
- ▶ S does not have to be a set of numbers, nor linearly ordered, nor ordered

A theory of maps with abstract domain

- ▶ **MapAD**: theory of maps with abstract domain
- ▶ **Store at inadmissible index i makes i admissible**:
$$\forall a, j, i, v. \text{Adm}(j, \text{len}(\text{store}(a, i, v))) \leftrightarrow (\text{Adm}(j, \text{len}(a)) \vee j \simeq i)$$
- ▶ **Store does not change length if the index is admissible**:
$$\forall a, i, v. \text{Adm}(i, \text{len}(a)) \rightarrow \text{len}(\text{store}(a, i, v)) \simeq \text{len}(a)$$
- ▶ **Select-over-store** axioms:
 - ▶ Restored: $\forall a, v, i. \text{select}(\text{store}(a, i, v), i) \simeq v$
 - ▶ $\forall a, v, i, j. i \not\simeq j \rightarrow \text{select}(\text{store}(a, i, v), j) \simeq \text{select}(a, j)$
- ▶ **Extensionality** unchanged: $\forall a, b. [\text{len}(a) \simeq \text{len}(b) \wedge (\forall i. \text{Adm}(i, \text{len}(a)) \rightarrow \text{select}(a, i) \simeq \text{select}(b, i))] \rightarrow a \simeq b$

A theory of vectors (dynamic arrays) with abstract domain

- ▶ **VecAD**: theory of vectors with abstract domain
- ▶ Store at an inadmissible index i makes i and the indices smaller than i admissible:
$$\forall a, j, i, v. \text{Adm}(j, \text{len}(\text{store}(a, i, v))) \leftrightarrow (\text{Adm}(j, \text{len}(a)) \vee j \leq i)$$
- ▶ Everything else as in **MapAD**
except for adding to the signature an ordering $<$ on indices
(does not have to be linear)
- ▶ **Dynamic** data structures axiomatized for the first time:
The theories of sequences do not do it

Theory modules for ArrAD, MapAD, VecAD

- ▶ From **axioms** to **inference rules**, e.g.:
 - ▶ $a \simeq b \vdash \text{len}(a) \simeq \text{len}(b)$
 - ▶ For **ArrAD**:
 $b \simeq \text{store}(a, i, v), \text{len}(b) \not\simeq \text{len}(a) \vdash \perp$
 - ▶ For **MapAD** and **VecAD**:
 $\text{len}(a) \simeq n, \text{Adm}(i, n), b \simeq \text{store}(a, i, v), \text{len}(b) \not\simeq \text{len}(a) \vdash \perp$
- ▶ Some rules generate $\perp$ (**conflict detection**) others do not:
balancing **finite local basis design** and **completeness**
- ▶ A **finite local basis** for **ArrAD**, **MapAD**, **VecAD**

Interpretation of arrays with abstract domain

Interpretation of arrays:

- ▶ An array: a function from indices to values
- ▶ Sort of arrays: an **updatable function set** X :
 $f \in X$ and g differs from f at finitely many indices: $g \in X$

Interpretation of **arrays with abstract domain**:

- ▶ An array of length n : a function from the set I_n of admissible indices (for n) to values
- ▶ Sort of arrays: a **collection of updatable function sets** $(X_n)_n$ one for each n in the interpretation of the sort L of lengths

Interpretation of maps with abstract domain

Sort of maps:

the collection $(X_n)_n$ must be **incrementally updatable**:

- ▶ if function f is in X_n and
- ▶ g is the function that maps i to v and is identical to f otherwise, then
- ▶ $\exists m$ such that $g \in X_m$ and
 - ▶ Either $m = n$ (store at an admissible i)
 - ▶ Or $I_m = I_n \cup \{i\}$
(store at an inadmissible index i which is admissible in the resulting map)

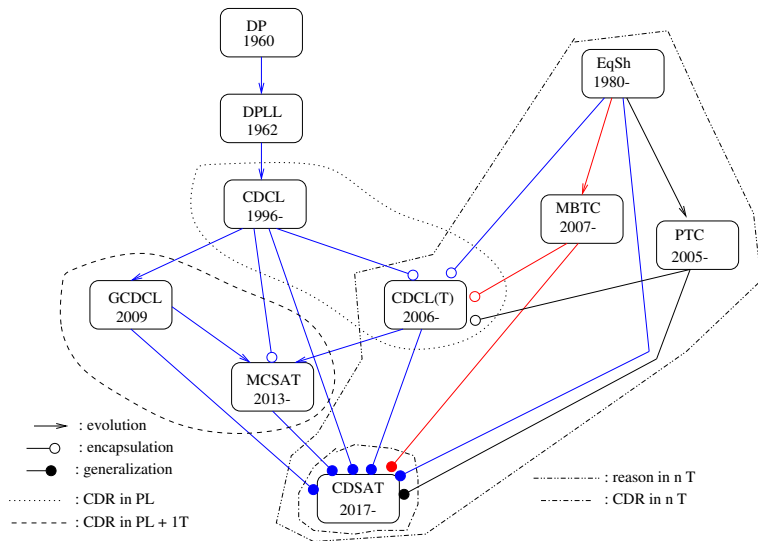
Interpretation of vectors with abstract domain

Sort of vectors:

the collection $(X_n)_n$ must be **extensibly updatable**:

- ▶ if function f is in X_n and
- ▶ g is the function that maps i to v and is identical to f otherwise, then
- ▶ $\exists m$ such that $g \in X_m$ and
 - ▶ Either $m = n$ (store at an admissible i)
 - ▶ Or $I_m = I_n \cup \{j \mid j \leq i\}$
(store at an inadmissible index i which is admissible in the resulting vector together with the smaller indices)

The big picture: more theory combination



Model-based theory combination (MBTC)

[de Moura, Bjørner: SMT 2007]

- ▶ Variant of equality sharing in CDCL($\mathcal{T}$)
- ▶ Only for $\mathcal{T}_k$ such that the $\mathcal{T}_k$ -procedure builds a candidate model $\mathcal{M}_k$ (e.g., LRA and Simplex)
- ▶ Idea: share $u \simeq v$ if $\mathcal{M}_k \models_{\mathcal{T}_k} (u \simeq v)$
- ▶ Motivation: enumerating $u \simeq v$ s.t. $\mathcal{M}_k \models_{\mathcal{T}_k} (u \simeq v)$ less expensive than enumerating $u \simeq v$ true in all $\mathcal{T}_k$ -models consistent with the current Boolean candidate model

Model-based theory combination (MBTC)

- ▶ Propositional abstraction of $u \simeq v$ is posted on the CDCL($\mathcal{T}$) trail by a **decision**
- ▶ If a **conflict** ensues, the decision is undone by CDCL backjumping and the $\mathcal{T}_k$ -procedure amends $\mathcal{M}_k$
- ▶ Terms u and v **cannot be new**
- ▶ $\mathcal{M}_k$ and conflict-driven updates remain inside the **black-box** $\mathcal{T}_k$ -procedure
- ▶ Useful to accelerate reaching **sat**

CDSAT generalizes MBTC

- ▶ All theory modules cooperate as **peers** to build a model for the union of the theories on the **shared trail**
- ▶ A theory module $\mathcal{I}_k$ can build a partial $\mathcal{T}_k$ -model $\mathcal{M}_k$ **publicly** on the **shared trail** by **first-order decisions**
- ▶ $\mathcal{I}_k$ can **deduce** an equality $u \simeq v$ that follows from assignments in $\mathcal{M}_k$, but is not a logical consequence of either the input or the Boolean contents of the trail because CDSAT modules **deduce** from **first-order assignments**

CDSAT generalizes MBTC

- ▶ If a conflict ensues, $u \simeq v$ and the **first-order decisions** from which it depends will be undone by **undoclear** or **undodecide** amending $\mathcal{M}_k$
- ▶ MBTC does it with a Boolean decision, because $\text{CDCL}(\mathcal{T})$ deduces only $\mathcal{T}$ -valid lemmas
- ▶ CDSAT opens the black-box procedures and achieves by **theory inferences** and **first-order decisions** on the **shared trail** what in other methods is hidden in the theory procedure

CDCL($\mathcal{T}$) with splitting on demand

- ▶ Entrust to CDCL the reasoning about **any** disjunction produced by the procedure of a non-convex theory
- ▶ The atoms in the clause must come from the existing clause set (i.e., input + equalities btw shared variables) or a **literal-generating function** (for termination)
- ▶ CDSAT: terms **inferred** by theory module **inferences** must come from **finite global basis** (for termination)
- ▶ CDSAT **decisions** are restricted by **acceptability** (which prevents $t \leftarrow c_1, \dots, t \leftarrow c_n, t \leftarrow c_{n+1}, \dots$) not by the **global basis**

CDCL($\mathcal{T}$) with splitting on demand

A first-order decision $t \leftarrow c$ in CDSAT cannot be mimicked by generating $t \simeq c \vee t \not\simeq c$ and then splitting it on demand, because

- ▶ c is not a term, it is a value coming from a theory extension (e.g., any rational number)
- ▶ No **literal-generating function** can produce $t \simeq c$ for an arbitrary c , because it would violate finiteness

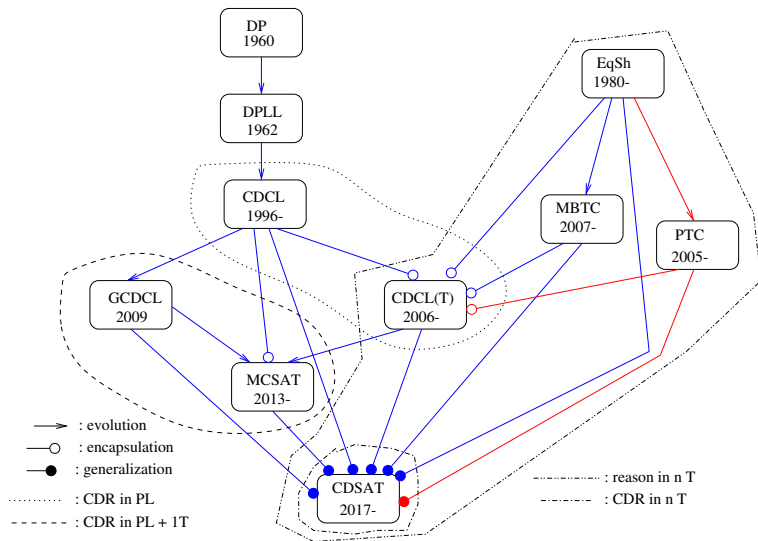
CDCL($\mathcal{T}$) with splitting on demand

If a conflict-driven $\mathcal{T}_k$ -procedure were integrated in CDCL($\mathcal{T}$):

- ▶ It'd be treated as a **black-box**
- ▶ Its $\mathcal{T}_k$ -specific conflict-driven mechanisms, namely
 - ▶ First-order $\mathcal{T}_k$ -assignments, and
 - ▶ New atoms generated by $\mathcal{T}_k$ -inferences

would remain private, would not be exported to the CDCL($\mathcal{T}$) trail and could not guide the global search

The big picture: more theory combination



Polite theory combination (PTC)

[Jovanović, Barrett: LPAR 2010] [Toledo, Przybicki, Zohar: CADE 2025]

- ▶ Variant of equality sharing in CDCL($\mathcal{T}$)
- ▶ Equality sharing requires the theories to be **stably infinite**
- ▶ PTC allows $\mathcal{T}_1$ **not stably infinite**, but $\mathcal{T}_2$ satisfies stronger cardinality requirements: **strongly polite**
- ▶ PTC combines theories by combining procedures
- ▶ Procedures combined as **black-boxes**
- ▶ Completeness approach like equality sharing: hypotheses on theories + combination theorem

CDSAT requires neither **stable infiniteness** nor **strong politeness**

CDSAT and agreement on cardinalities of shared sorts

- ▶ CDSAT requires that there exists **leading theory**, say $\mathcal{T}_1$, that
 - ▶ Has all sorts in the theory union
 - ▶ Has all cardinality constraints aggregated and enforced by $\mathcal{T}_1$ -module inferences
- ▶ Every $\mathcal{T}_k$ ($k \neq 1$) has to agree with $\mathcal{T}_1$ on what's shared: any two $\mathcal{T}_k$ and $\mathcal{T}_j$ ($k \neq j$) agree
- ▶ Agreement guaranteed by theory modules **completeness** requirements
- ▶ Different approach to **completeness**:
 - ▶ $\mathcal{T}_1$ -module **complete**
 - ▶ $\mathcal{T}_k$ -module ($k \neq 1$) **leading-theory-complete**

Examples about cardinalities of shared sorts

1. All theories **stably infinite**: $\mathcal{T}_1$ is fictional $\mathcal{T}_{\mathbb{N}}$ that interprets all sorts (except prop) as having the cardinality of $\mathbb{N}$
2. **At-most- m** cardinality constraint on sort s :
$$\forall x_0, \dots, \forall x_m. \bigvee_{0 \leq i \neq k \leq m} x_i \simeq_s x_k$$

 $x_0, \dots, x_m$: $m + 1$ distinct variables of sort s
Inference rule in the $\mathcal{T}_1$ -module:
$$\bigwedge_{0 \leq i \neq k \leq m} u_i \not\simeq_s u_k \vdash_{\mathcal{T}_1} \perp$$

 $u_0, \dots, u_m$: any $m + 1$ distinct terms of sort s
3. Aggregation: if $\mathcal{T}_2$ says **at-most- m** and $\mathcal{T}_3$ says **at-most- p** ,
leading theory $\mathcal{T}_1$ says **at-most- $\min(m, p)$**

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Author: MPB

A forthcoming book: stay tuned

- **Conflict-Driven Procedures for Automated Reasoning.**
Research monograph, Springer Series on Theories and Applications of Computability, in preparation, and expected June 2027.

Author: MPB

Thank you!